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(54) **APERTURE-METASURFACE AND HYBRID REFRACTIVE-METASURFACE IMAGING SYSTEMS**

(71) Applicant: **Metallenz, Inc.**, Boston, MA (US)

(72) Inventors: **Robert C. Devlin**, Stoneham, MA (US); **John Graff**, Swampscott, MA (US)

(73) Assignee: **Metallenz, Inc.**, Boston, MA (US)

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,877,034 A 4/1975 Nelson
4,777,116 A 10/1988 Kawatsuki et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CA 3006173 A1 6/2017
CA 3020261 A1 10/2017
(Continued)

OTHER PUBLICATIONS

Engelberg et al., Near-IR Wide Field-of-View Huygens Metalens for Outdoor Imaging Applications, 2019 (Year: 2019).
(Continued)

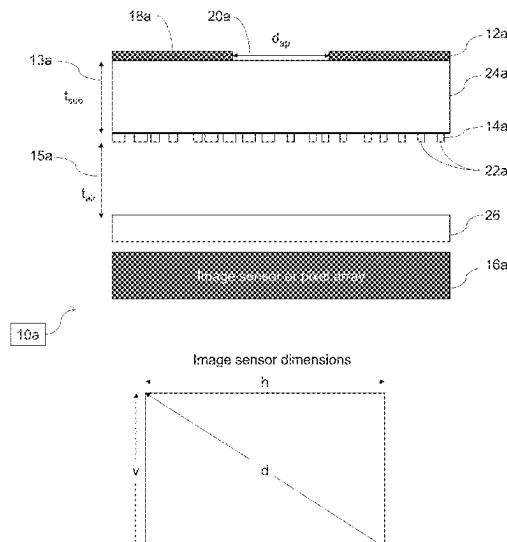
Primary Examiner — Ephrem Z Mebrahtu

(74) *Attorney, Agent, or Firm* — KPPB LLP

(57) **ABSTRACT**

Hybrid imaging systems incorporating conventional optical elements and metasurface elements with light sources and/or detectors, and methods of the manufacture and operation of such optical arrangements are provided. Systems and methods describe the integration of apertures with metasurface elements and refractive optics with metasurface elements in illumination sources and sensors.

18 Claims, 20 Drawing Sheets



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(56) References Cited

U.S. PATENT DOCUMENTS

4,856,899	A	8/1989	Iwaoka et al.	8,848,273	B2	9/2014	Yu et al.
5,085,496	A	2/1992	Yoshida et al.	8,876,289	B2	11/2014	Dorransoro Diaz et al.
5,245,466	A	9/1993	Burns et al.	8,908,149	B2	12/2014	Freimann
5,337,146	A	8/1994	Azzam	8,912,973	B2	12/2014	Werner et al.
5,452,126	A	9/1995	Johnson	8,981,337	B1	3/2015	Burckel et al.
5,620,792	A	4/1997	Challener, IV	9,007,451	B2	4/2015	Rogers et al.
6,097,856	A	8/2000	Hammond, Jr.	9,116,302	B2	8/2015	Mccarthy et al.
6,643,065	B1	11/2003	Silberman	9,151,891	B2	10/2015	Ma et al.
6,669,803	B1	12/2003	Kathman et al.	9,212,899	B2	12/2015	Johnson et al.
6,731,839	B2	5/2004	Bhagavatula et al.	9,298,060	B2	3/2016	Shen et al.
6,825,986	B2	11/2004	Ashkinazy et al.	9,309,274	B2	4/2016	Van Der et al.
6,834,027	B1	12/2004	Sakaguchi et al.	9,310,535	B1	4/2016	Greiner et al.
6,924,457	B2	8/2005	Koyama et al.	9,329,484	B1	5/2016	Markle et al.
6,927,922	B2	8/2005	George et al.	9,330,704	B2	5/2016	Nishimura et al.
7,057,151	B2	6/2006	Lezec et al.	9,367,036	B2	6/2016	Pyun et al.
7,061,612	B2	6/2006	Johnston	9,369,621	B2	6/2016	Malone et al.
7,061,693	B2	6/2006	Zalevsky	9,391,700	B1	7/2016	Bruce et al.
7,171,078	B2	1/2007	Sasaki et al.	9,392,153	B2	7/2016	Myhre et al.
7,171,084	B2	1/2007	Izumi et al.	9,411,103	B2	8/2016	Astratov
7,186,969	B2	3/2007	Altendorf et al.	9,482,796	B2	11/2016	Arbabi et al.
7,241,988	B2	7/2007	Gruber et al.	9,500,771	B2	11/2016	Liu et al.
7,324,210	B2	1/2008	De Groot et al.	9,536,362	B2	1/2017	Sarwar et al.
7,327,468	B2	2/2008	Maznev et al.	9,553,423	B2	1/2017	Chen et al.
7,402,131	B2	7/2008	Mueth et al.	9,557,585	B1	1/2017	Yap et al.
7,450,618	B2	11/2008	Dantus et al.	9,606,415	B2	3/2017	Zheludev et al.
7,547,874	B2	6/2009	Liang	9,609,190	B2	3/2017	Lee et al.
7,561,264	B2	7/2009	Treado et al.	9,704,250	B1	7/2017	Shah et al.
7,576,899	B2	8/2009	Kanesaka et al.	9,739,918	B2	8/2017	Arbabi et al.
7,679,830	B2	3/2010	Dowski, Jr.	9,766,463	B2	9/2017	Border et al.
7,684,097	B2	3/2010	Fukumoto et al.	9,778,404	B2	10/2017	Divliansky et al.
7,773,307	B2	8/2010	Shih	9,825,074	B2	11/2017	Tian et al.
7,800,683	B2	9/2010	Zalevsky et al.	9,829,700	B2	11/2017	Parent et al.
7,812,295	B2	10/2010	Zalevsky et al.	9,835,870	B2	12/2017	Astratov et al.
7,928,900	B2	4/2011	Fuller et al.	9,836,122	B2	12/2017	Border
7,929,220	B2	4/2011	Sayag	9,869,580	B2	1/2018	Grossinger et al.
7,965,607	B2	6/2011	Fukumoto et al.	9,880,377	B1	1/2018	Safrani et al.
8,009,358	B2	8/2011	Zalevsky et al.	9,885,859	B2	2/2018	Harris
8,040,604	B2	10/2011	Zalevsky et al.	9,891,393	B2	2/2018	Reece
8,107,705	B2	1/2012	Dowski, Jr. et al.	9,939,129	B2	4/2018	Byrnes et al.
8,152,307	B2	4/2012	Duelli et al.	9,947,118	B2	4/2018	Khare et al.
8,169,703	B1	5/2012	Mossberg et al.	9,952,096	B2	4/2018	Kats et al.
8,192,022	B2	6/2012	Zalevsky	9,958,251	B1	5/2018	Brock et al.
8,212,866	B2	7/2012	Lemmer et al.	9,967,541	B2	5/2018	Piestun
8,318,386	B2	11/2012	Kobrin	9,978,801	B2	5/2018	Park et al.
8,328,396	B2	12/2012	Capasso et al.	9,989,680	B2	6/2018	Arbabi et al.
8,351,048	B2	1/2013	Millerd	9,992,474	B2	6/2018	Grunnet-Jepsen et al.
8,351,120	B2	1/2013	Deng et al.	9,995,859	B2	6/2018	Kamali et al.
8,390,932	B2	3/2013	Jia et al.	9,995,930	B2 *	6/2018	Arbabi G02B 5/0294
8,400,494	B2	3/2013	Zalevsky et al.	10,007,118	B2	6/2018	Border
8,430,513	B2	4/2013	Chang et al.	10,054,859	B2	8/2018	Ye et al.
8,451,368	B2	5/2013	Sung et al.	10,084,239	B2	9/2018	Shaver et al.
8,472,797	B2	6/2013	Ok et al.	10,108,085	B2	10/2018	Peters et al.
8,481,948	B2	7/2013	Frach et al.	10,126,466	B2	11/2018	Lin et al.
8,558,873	B2	10/2013	Mceldowney	10,132,465	B2	11/2018	Byrnes et al.
8,587,474	B2	11/2013	Fuller et al.	10,149,612	B2	12/2018	Muyo et al.
8,649,631	B2	2/2014	Islam et al.	10,155,846	B2	12/2018	Fuji et al.
8,681,428	B1	3/2014	Brown	10,234,383	B2	3/2019	Wang et al.
8,687,040	B2	4/2014	Silveira	10,254,454	B2	4/2019	Klug et al.
8,716,677	B2	5/2014	Cui	10,267,957	B2	4/2019	Kamali et al.
8,734,033	B2	5/2014	Walters et al.	10,310,148	B2	6/2019	Stewart et al.
8,743,923	B2	6/2014	Geske et al.	10,310,387	B2	6/2019	Palmer et al.
8,816,460	B2	8/2014	Kalevo et al.	10,315,951	B2	6/2019	Toussaint et al.
				10,317,667	B2	6/2019	Waller et al.
				10,324,314	B2	6/2019	Czaplewski et al.
				10,338,275	B1	7/2019	Acosta et al.
				10,341,640	B2	7/2019	Shechtman et al.
				10,345,246	B2	7/2019	Tian et al.
				10,345,519	B1	7/2019	Miller et al.
				10,365,416	B2	7/2019	Zhan et al.
				10,371,936	B2	8/2019	Didomenico
				10,386,620	B2	8/2019	Astratov et al.
				10,388,805	B2	8/2019	Engel et al.
				10,402,993	B2	9/2019	Han et al.
				10,408,416	B2	9/2019	Khorasaninejad et al.
				10,408,419	B2	9/2019	Aieta et al.
				10,416,565	B2	9/2019	Ahmed et al.
				10,435,814	B2	10/2019	Plummer et al.
				10,440,244	B2	10/2019	Rosenblatt et al.
				10,440,300	B2	10/2019	Rephaeli et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

10,466,394	B2	11/2019	Lin et al.	2011/0012807	A1	1/2011	Sorvala
10,468,447	B2	11/2019	Akselrod et al.	2011/0019180	A1	1/2011	Kruglick
10,481,317	B2	11/2019	Peroz et al.	2011/0102877	A1	5/2011	Parriaux
10,514,296	B2	12/2019	Han et al.	2011/0149251	A1	6/2011	Duelli
10,527,832	B2	1/2020	Schwab et al.	2011/0187577	A1	8/2011	Fuller et al.
10,527,851	B2	1/2020	Lin et al.	2011/0261441	A1	10/2011	Zheludev et al.
10,536,688	B2	1/2020	Haas et al.	2012/0008133	A1	1/2012	Silny et al.
10,539,723	B2	1/2020	Iazikov et al.	2012/0068347	A1	3/2012	Isobayashi et al.
10,545,323	B2	1/2020	Schwab et al.	2012/0092735	A1	4/2012	Futterer et al.
10,591,643	B2	3/2020	Lin et al.	2012/0140235	A1	6/2012	Lee et al.
10,670,782	B2	6/2020	Arbabi et al.	2012/0258407	A1	10/2012	Sirat
10,725,290	B2	7/2020	Fan et al.	2012/0269483	A1	10/2012	Mossberg et al.
10,795,168	B2	10/2020	Riley, Jr. et al.	2012/0293854	A1	11/2012	Zheludev et al.
10,816,704	B2	10/2020	Arbabi et al.	2012/0327666	A1	12/2012	Liu et al.
10,816,815	B2	10/2020	Aieta et al.	2012/0328240	A1	12/2012	Ma et al.
10,915,737	B2	2/2021	Hu et al.	2013/0016030	A1	1/2013	Liu et al.
10,916,060	B2	2/2021	West et al.	2013/0032949	A1	2/2013	Lin et al.
11,092,717	B2	8/2021	Capasso et al.	2013/0037873	A1	2/2013	Suzuki et al.
11,169,311	B2	11/2021	Rubin et al.	2013/0050285	A1	2/2013	Takahashi et al.
11,231,544	B2	1/2022	Lin et al.	2013/0058071	A1	3/2013	Ben
11,298,052	B2	4/2022	Palikaras et al.	2013/0194537	A1	8/2013	Mao et al.
11,353,626	B2	6/2022	You et al.	2013/0194787	A1	8/2013	Geske et al.
11,366,296	B2	6/2022	Devlin et al.	2013/0208332	A1	8/2013	Yu et al.
11,385,104	B2	7/2022	Yao et al.	2014/0009823	A1	1/2014	Park et al.
11,385,516	B2	7/2022	Didomenico	2014/0043846	A1	2/2014	Yang et al.
11,578,968	B1	2/2023	Capasso et al.	2014/0085693	A1	3/2014	Mosallaei et al.
11,579,456	B2	2/2023	Riley et al.	2014/0210835	A1	7/2014	Hong et al.
11,604,364	B2	3/2023	Rubin et al.	2014/0277433	A1	9/2014	Pugh et al.
11,733,535	B2	8/2023	Aieta et al.	2015/0011073	A1	1/2015	Lei et al.
11,815,668	B2	11/2023	Devlin et al.	2015/0017466	A1	1/2015	Ayon et al.
11,835,680	B2	12/2023	Groeve et al.	2015/0018500	A1	1/2015	Gerber et al.
11,835,681	B2	12/2023	Lin et al.	2015/0055745	A1	2/2015	Holzner et al.
11,867,937	B2	1/2024	Rubin et al.	2015/0068599	A1	3/2015	Chou et al.
11,978,752	B2	5/2024	Devlin et al.	2015/0090862	A1	4/2015	Matsui et al.
11,988,844	B2	5/2024	Riley, Jr. et al.	2015/0092139	A1	4/2015	Eguchi
2002/0048727	A1	4/2002	Zhou et al.	2015/0098002	A1	4/2015	Wang
2002/0118903	A1	8/2002	Cottrell et al.	2015/0116721	A1	4/2015	Kats et al.
2002/0181126	A1	12/2002	Nishioka	2015/0125111	A1	5/2015	Orcutt et al.
2003/0077983	A1	4/2003	Hagan et al.	2015/0185413	A1	7/2015	Greiner et al.
2003/0107787	A1	6/2003	Bablumyan	2015/0219497	A1	8/2015	Johs et al.
2004/0173738	A1	9/2004	Mizuno	2015/0219806	A1	8/2015	Arbabi et al.
2004/0184752	A1	9/2004	Aoki et al.	2015/0241608	A1	8/2015	Shian et al.
2004/0190116	A1	9/2004	Lezec et al.	2015/0316717	A1	11/2015	Astratov
2004/0258128	A1	12/2004	Johs et al.	2016/0025914	A1	1/2016	Brongersma et al.
2005/0151698	A1	7/2005	Mohamadi	2016/0037146	A1	2/2016	McGrew
2005/0161589	A1	7/2005	Kim et al.	2016/0077261	A1	3/2016	Arbabi et al.
2005/0211665	A1	9/2005	Gao et al.	2016/0133762	A1	5/2016	Blasco Claret et al.
2005/0220162	A1	10/2005	Nakamura	2016/0161826	A1	6/2016	Shen et al.
2005/0239003	A1	10/2005	Chiodini et al.	2016/0195705	A1	7/2016	Betzig et al.
2006/0042322	A1	3/2006	Mendoza et al.	2016/0253551	A1	9/2016	Pezzaniti et al.
2006/0127829	A1	6/2006	Deng et al.	2016/0254638	A1	9/2016	Chen et al.
2007/0024975	A1	2/2007	McGrew	2016/0276979	A1	9/2016	Shaver et al.
2007/0026585	A1	2/2007	Wong et al.	2016/0299337	A1	10/2016	Arbabi et al.
2007/0030870	A1	2/2007	Bour et al.	2016/0299426	A1	10/2016	Gates et al.
2007/0273957	A1	11/2007	Zalevsky et al.	2016/0306079	A1	10/2016	Arbabi et al.
2008/0014632	A1	1/2008	Cunningham et al.	2016/0306157	A1	10/2016	Rho et al.
2009/0122175	A1*	5/2009	Yamagata	2016/0318067	A1	11/2016	Banerjee et al.
			H10F 39/8063	2016/0331457	A1	11/2016	Varghese et al.
			348/335	2016/0341859	A1	11/2016	Shvets et al.
2009/0128908	A1	5/2009	Nakazawa et al.	2016/0359235	A1	12/2016	Driscoll et al.
2009/0135086	A1	5/2009	Fuller et al.	2016/0361002	A1	12/2016	Palikaras et al.
2009/0230333	A1	9/2009	Eleftheriades	2016/0370568	A1	12/2016	Toussaint et al.
2009/0296223	A1	12/2009	Werner et al.	2017/0003169	A1	1/2017	Shaltout et al.
2010/0033701	A1	2/2010	Lee et al.	2017/0010466	A1	1/2017	Klug et al.
2010/0055621	A1	3/2010	Hatakeyama et al.	2017/0030773	A1	2/2017	Han et al.
2010/0072170	A1	3/2010	Wu et al.	2017/0038574	A1	2/2017	Zhuang et al.
2010/0091224	A1	4/2010	Cho et al.	2017/0045652	A1	2/2017	Arbabi et al.
2010/0110430	A1	5/2010	Ebbesen et al.	2017/0082263	A1*	3/2017	Byrnes
2010/0110433	A1	5/2010	Nedelcu et al.	2017/0090221	A1	3/2017	Atwater
2010/0134869	A1	6/2010	Bernet et al.	2017/0121843	A1	5/2017	Plummer et al.
2010/0177164	A1	7/2010	Zalevsky et al.	2017/0125911	A1	5/2017	Alu et al.
2010/0187658	A1	7/2010	Wei	2017/0131460	A1	5/2017	Lin et al.
2010/0226134	A1	9/2010	Capasso et al.	2017/0146806	A1	5/2017	Lin et al.
2010/0232017	A1	9/2010	Mccarthy et al.	2017/0176758	A1	6/2017	Lerner et al.
2010/0255428	A1	10/2010	Chen et al.	2017/0186166	A1	6/2017	Grunnet-Jepsen et al.
2010/0259804	A1	10/2010	Buschbeck et al.	2017/0201658	A1	7/2017	Rosenblatt et al.
				2017/0212285	A1	7/2017	Arbabi et al.
				2017/0235162	A1	8/2017	Shaltout et al.
				2017/0250577	A1	8/2017	Ho et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2017/0293141	A1	10/2017	Schowengerdt et al.		2020/0124866	A1 *	4/2020	Camayd-Munoz	G02B 1/002
2017/0299784	A1	10/2017	Mikkelsen et al.		2020/0150311	A1 *	5/2020	Zhang	G02B 1/002
2017/0310907	A1 *	10/2017	Wang	G02B 1/002	2020/0225386	A1	7/2020	Tsai et al.	
2017/0322418	A1	11/2017	Lin et al.		2020/0236315	A1	7/2020	Kimura	
2017/0329201	A1	11/2017	Arnold		2020/0249429	A1	8/2020	Han et al.	
2017/0374352	A1	12/2017	Horesh		2020/0271941	A1	8/2020	Riley, Jr. et al.	
2018/0035101	A1	2/2018	Osterhout		2020/0272100	A1	8/2020	Yu et al.	
2018/0044234	A1	2/2018	Hokansson et al.		2020/0284960	A1	9/2020	Ellenbogen et al.	
2018/0045953	A1	2/2018	Fan et al.		2020/0355913	A1	11/2020	Park et al.	
2018/0052276	A1	2/2018	Klienman et al.		2020/0404183	A1 *	12/2020	Kimura	H04N 25/61
2018/0052320	A1	2/2018	Curtis et al.		2021/0010928	A1	1/2021	Acher et al.	
2018/0095193	A1 *	4/2018	Wang	G02B 3/0056	2021/0028215	A1	1/2021	Devlin et al.	
2018/0107015	A1	4/2018	Dümpelmann et al.		2021/0048569	A1	2/2021	Rubin et al.	
2018/0109002	A1	4/2018	Foo		2021/0109364	A1	4/2021	Aieta et al.	
2018/0129866	A1	5/2018	Hicks et al.		2021/0112201	A1	4/2021	Cho et al.	
2018/0172988	A1	6/2018	Ahmed et al.		2021/0149081	A1	5/2021	Groeveer et al.	
2018/0184065	A1	6/2018	Zhao et al.		2021/0190593	A1	6/2021	Yao et al.	
2018/0216797	A1	8/2018	Khorasaninejad et al.		2021/0200992	A1	7/2021	Padmanabhan et al.	
2018/0217395	A1	8/2018	Lin et al.		2021/0208469	A1	7/2021	Didomenico	
2018/0231700	A1	8/2018	Ahmed et al.		2021/0263329	A1	8/2021	Latawiec	
2018/0231702	A1	8/2018	Lin et al.		2021/0286188	A1	9/2021	Rubin et al.	
2018/0236596	A1	8/2018	Ihleemann et al.		2021/0288095	A1	9/2021	Delga et al.	
2018/0246262	A1	8/2018	Zhan et al.		2021/0302763	A1	9/2021	Yao et al.	
2018/0248268	A1	8/2018	Shvets et al.		2021/0311588	A1	10/2021	Han et al.	
2018/0252857	A1	9/2018	Glik et al.		2021/0318466	A1	10/2021	Uenoyama et al.	
2018/0259700	A1	9/2018	Khorasaninejad et al.		2021/0333150	A1	10/2021	Mceldowney et al.	
2018/0259757	A1	9/2018	Urzhumov		2022/0050294	A1	2/2022	Fermigier et al.	
2018/0267605	A1	9/2018	Border		2022/0091428	A1	3/2022	Riley, Jr. et al.	
2018/0274750	A1	9/2018	Byrnes et al.		2022/0103753	A1 *	3/2022	Kojima	H04N 25/61
2018/0292644	A1	10/2018	Kamali et al.		2022/0107263	A1 *	4/2022	Biesinger	G01J 3/0205
2018/0299595	A1	10/2018	Arbabi et al.		2022/0196480	A1 *	6/2022	Ang	G01J 5/045
2018/0314130	A1	11/2018	Joo et al.		2022/0206186	A1 *	6/2022	Chen	G02B 1/002
2018/0341090	A1	11/2018	Devlin et al.		2022/0206205	A1	6/2022	Rubin et al.	
2018/0364158	A1	12/2018	Wang et al.		2022/0214219	A1	7/2022	Faraon et al.	
2019/0003892	A1	1/2019	Aieta et al.		2022/0244442	A1	8/2022	Rubin et al.	
2019/0025463	A1	1/2019	She et al.		2022/0283411	A1	9/2022	Devlin et al.	
2019/0025477	A1	1/2019	She et al.		2023/0194883	A1	6/2023	Riley et al.	
2019/0041642	A1	2/2019	Haddick et al.		2023/0196842	A1	6/2023	Devlin et al.	
2019/0041660	A1	2/2019	Ahmed		2023/0208104	A1	6/2023	Tamagnone et al.	
2019/0041736	A1 *	2/2019	Grunnet-Jepsen ...	G03B 21/005	2023/0280498	A1	9/2023	Altuzarra et al.	
2019/0044003	A1	2/2019	Heck et al.		2023/0288716	A1	9/2023	Rubin et al.	
2019/0049632	A1	2/2019	Shin et al.		2023/0314827	A1	10/2023	Devlin et al.	
2019/0049732	A1	2/2019	Lee et al.		2023/0318261	A1	10/2023	Tamagnone et al.	
2019/0057512	A1	2/2019	Han et al.		2024/0210246	A1	6/2024	Rubin et al.	
2019/0064532	A1	2/2019	Riley, Jr. et al.		2024/0234461	A1	7/2024	Devlin et al.	
2019/0079321	A1	3/2019	Wu et al.						
2019/0086579	A1	3/2019	Kim et al.						
2019/0086683	A1	3/2019	Aieta et al.						
2019/0101448	A1	4/2019	Lee et al.						
2019/0113775	A1	4/2019	Jang et al.						
2019/0120817	A1	4/2019	Anderson						
2019/0121004	A1	4/2019	Ahmed et al.						
2019/0137075	A1	5/2019	Aieta et al.						
2019/0137762	A1	5/2019	Hu						
2019/0137793	A1	5/2019	Luo et al.						
2019/0154877	A1 *	5/2019	Capasso	G02B 5/18					
2019/0155302	A1	5/2019	Lukierski et al.						
2019/0157830	A1	5/2019	Tong et al.						
2019/0162592	A1 *	5/2019	Khorasaninejad	G01J 3/00					
2019/0170655	A1	6/2019	Smith						
2019/0191144	A1	6/2019	Arbabi et al.						
2019/0196068	A1	6/2019	Tsai et al.						
2019/0206136	A1	7/2019	West et al.						
2019/0219835	A1	7/2019	Skinner et al.						
2019/0235139	A1	8/2019	Chen et al.						
2019/0250107	A1	8/2019	Sreenivasan et al.						
2019/0369401	A1	12/2019	Rolland et al.						
2019/0377084	A1	12/2019	Sleasman et al.						
2019/0386749	A1	12/2019	Lezec et al.						
2019/0391378	A1	12/2019	Eichelkraut et al.						
2020/0025888	A1	1/2020	Jang et al.						
2020/0052027	A1	2/2020	Arbabi et al.						
2020/0076163	A1	3/2020	Han et al.						
2020/0083666	A1	3/2020	Fallahi et al.						
2020/0096672	A1	3/2020	Yu et al.						

FOREIGN PATENT DOCUMENTS

CA	3064764	A1	11/2018
CA	217639612	U	10/2022
CA	217820840	U	11/2022
CN	1044991	A	8/1990
CN	101158727	A	4/2008
CN	101164147	A	4/2008
CN	100476504	C	4/2009
CN	101546002	A	9/2009
CN	101681095	A	3/2010
CN	101510013	B	6/2010
CN	101510012	B	8/2010
CN	101510011	B	9/2010
CN	101241173	B	8/2011
CN	202854395	U	4/2013
CN	103092049	A	5/2013
CN	203799117	U	8/2014
CN	104067171	A	9/2014
CN	104374745	A	2/2015
CN	204422813	U	6/2015
CN	104932043	A	9/2015
CN	104956491	A	9/2015
CN	204719330	U	10/2015
CN	105068396	A	11/2015
CN	103869484	B	1/2016
CN	105278026	A	1/2016
CN	105278309	A	1/2016
CN	105655286	A	6/2016
CN	105676314	A	6/2016
CN	105917277	A	8/2016
CN	105974503	A	9/2016
CN	103257441	B	10/2016

(56)	References Cited		CN	114325886	A	4/2022
	FOREIGN PATENT DOCUMENTS		CN	114326163	A	4/2022
			CN	114354141	A	4/2022
			CN	114373825	A	4/2022
CN	205620619	U 10/2016	CN	114384612	A	4/2022
CN	104834079	B 4/2017	CN	114397092	A	4/2022
CN	106611699	A 5/2017	CN	114397718	A	4/2022
CN	104834089	B 6/2017	CN	114415386	A	4/2022
CN	106848555	A 6/2017	CN	216345776	U	4/2022
CN	106200276	B 10/2017	CN	216351311	U	4/2022
CN	104834088	B 12/2017	CN	216351591	U	4/2022
CN	105676314	B 1/2018	CN	216355281	U	4/2022
CN	107561857	A 1/2018	CN	216361353	U	4/2022
CN	108089325	A 5/2018	CN	111316138	B	5/2022
CN	108291983	A 7/2018	CN	114488365	A	5/2022
CN	207623619	U 7/2018	CN	114543993	A	5/2022
CN	106199997	B 8/2018	CN	114545367	A	5/2022
CN	108474869	A 8/2018	CN	114545370	A	5/2022
CN	108507542	A 9/2018	CN	114554062	A	5/2022
CN	207923075	U 9/2018	CN	114561266	A	5/2022
CN	108680544	A 10/2018	CN	216593224	U	5/2022
CN	108761779	A 11/2018	CN	216605227	U	5/2022
CN	109000692	A 12/2018	CN	216622749	U	5/2022
CN	208270846	U 12/2018	CN	114578642	A	6/2022
CN	109196387	A 1/2019	CN	114593689	A	6/2022
CN	208421387	U 1/2019	CN	114623960	A	6/2022
CN	106199956	B 2/2019	CN	114624878	A	6/2022
CN	109360139	A 2/2019	CN	114660683	A	6/2022
CN	106950195	B 5/2019	CN	114660780	A	6/2022
CN	106324832	B 7/2019	CN	114690387	A	7/2022
CN	106526730	B 7/2019	CN	114740631	A	7/2022
CN	106485761	B 8/2019	CN	114743714	A	7/2022
CN	110160685	A 8/2019	CN	114779437	U	7/2022
CN	110678773	A 1/2020	CN	216896898	U	7/2022
CN	108474869	B 6/2020	CN	216900930	U	7/2022
CN	111316138	A 6/2020	CN	216901121	U	7/2022
CN	111580190	A 8/2020	CN	216901165	U	7/2022
CN	111656707	A 9/2020	CN	216901317	U	7/2022
CN	111819489	A 10/2020	CN	216901952	U	7/2022
CN	213092332	U 4/2021	CN	216903719	U	7/2022
CN	113050295	A 6/2021	CN	216933177	U	7/2022
CN	113168022	A 7/2021	CN	217034311	U	7/2022
CN	110376665	B 8/2021	CN	217034418	U	7/2022
CN	213902664	U 8/2021	CN	217034466	U	7/2022
CN	213903843	U 8/2021	CN	114859446	A	8/2022
CN	214098104	U 8/2021	CN	114859447	A	8/2022
CN	113703080	A 11/2021	CN	114859570	A	8/2022
CN	111580190	B 12/2021	CN	114935741	A	8/2022
CN	113791524	A 12/2021	CN	217276608	U	8/2022
CN	113807312	A 12/2021	CN	217278911	U	8/2022
CN	113820839	A 12/2021	CN	217278915	U	8/2022
CN	113834568	A 12/2021	CN	217278989	U	8/2022
CN	113835227	A 12/2021	CN	217279003	U	8/2022
CN	113851573	A 12/2021	CN	217279087	U	8/2022
CN	215005942	U 12/2021	CN	217279110	U	8/2022
CN	215010478	U 12/2021	CN	217279168	U	8/2022
CN	110494771	B 1/2022	CN	217279244	U	8/2022
CN	113885106	A 1/2022	CN	217280797	U	8/2022
CN	113899451	A 1/2022	CN	217280851	U	8/2022
CN	113900078	A 1/2022	CN	217281621	U	8/2022
CN	113900162	A 1/2022	CN	217281623	U	8/2022
CN	113906320	A 1/2022	CN	114995038	A	9/2022
CN	113917574	A 1/2022	CN	115016099	A	9/2022
CN	113917578	A 1/2022	CN	115016150	A	9/2022
CN	113934004	A 1/2022	CN	115032766	A	9/2022
CN	113934005	A 1/2022	CN	115047432	A	9/2022
CN	113959984	A 1/2022	CN	115047548	A	9/2022
CN	114002707	A 2/2022	CN	115047653	A	9/2022
CN	114019589	A 2/2022	CN	115061114	A	9/2022
CN	114047632	A 2/2022	CN	115079415	A	9/2022
CN	114047637	A 2/2022	CN	115113174	A	9/2022
CN	114112058	A 3/2022	CN	217456368	U	9/2022
CN	114156168	A 3/2022	CN	217465697	U	9/2022
CN	114176492	A 3/2022	CN	217466052	U	9/2022
CN	215932365	U 3/2022	CN	217466667	U	9/2022
CN	114280704	A 4/2022	CN	217467162	U	9/2022
CN	114280716	A 4/2022	CN	217467176	U	9/2022
CN	114286953	A 4/2022	CN	217467177	U	9/2022
CN	114296180	A 4/2022	CN	217467226	U	9/2022

(56)	References Cited					
	FOREIGN PATENT DOCUMENTS					
CN	217467326	U	9/2022	CN	217821071	U 11/2022
CN	217467327	U	9/2022	CN	217821091	U 11/2022
CN	217467336	U	9/2022	CN	217821110	U 11/2022
CN	217467338	U	9/2022	CN	217821111	U 11/2022
CN	217467351	U	9/2022	CN	217821113	U 11/2022
CN	217467352	U	9/2022	CN	217821122	U 11/2022
CN	217467353	U	9/2022	CN	217821160	U 11/2022
CN	217467355	U	9/2022	CN	217821236	U 11/2022
CN	217467357	U	9/2022	CN	217821680	U 11/2022
CN	217467358	U	9/2022	CN	217821696	U 11/2022
CN	217467363	U	9/2022	CN	217822825	U 11/2022
CN	217467364	U	9/2022	CN	217823690	U 11/2022
CN	217467367	U	9/2022	CN	217825178	U 11/2022
CN	217467368	U	9/2022	CN	217885960	U 11/2022
CN	217467395	U	9/2022	CN	217902220	U 11/2022
CN	217467396	U	9/2022	CN	217902222	U 11/2022
CN	217467399	U	9/2022	CN	115421295	A 12/2022
CN	217467439	U	9/2022	CN	115453754	A 12/2022
CN	217467452	U	9/2022	CN	115524768	A 12/2022
CN	115164714	A	10/2022	CN	115524775	A 12/2022
CN	115166876	A	10/2022	CN	115524874	A 12/2022
CN	115166958	A	10/2022	CN	217981833	U 12/2022
CN	115185082	A	10/2022	CN	217981857	U 12/2022
CN	115211799	A	10/2022	CN	217981991	U 12/2022
CN	115236795	A	10/2022	CN	217981992	U 12/2022
CN	217639515	U	10/2022	CN	217982020	U 12/2022
CN	217639519	U	10/2022	CN	217982038	U 12/2022
CN	217639539	U	10/2022	CN	217982089	U 12/2022
CN	217639544	U	10/2022	CN	217982120	U 12/2022
CN	217639611	U	10/2022	CN	217983382	U 12/2022
CN	217639613	U	10/2022	CN	217984044	U 12/2022
CN	217639715	U	10/2022	CN	115812169	A 3/2023
CN	217639718	U	10/2022	CN	116745685	A 9/2023
CN	217639719	U	10/2022	DE	102007058558	A1 6/2009
CN	217639720	U	10/2022	DE	102009037629	A1 2/2011
CN	217639722	U	10/2022	DE	102012212753	A1 1/2014
CN	217639723	U	10/2022	DE	102015221985	A1 5/2017
CN	217639724	U	10/2022	DE	102016218996	A1 9/2017
CN	217639725	U	10/2022	DE	112018002811	T5 2/2020
CN	217639726	U	10/2022	DE	112018002670	T5 3/2020
CN	217639763	U	10/2022	EP	1251397	A2 10/2002
CN	217639765	U	10/2022	EP	1252623	B1 10/2004
CN	217639767	U	10/2022	EP	2631740	A2 8/2013
CN	217639768	U	10/2022	EP	2763519	A2 8/2014
CN	217639769	U	10/2022	EP	2338114	B1 3/2017
CN	217639770	U	10/2022	EP	3226042	A1 10/2017
CN	217639771	U	10/2022	EP	3353578	A1 8/2018
CN	217639772	U	10/2022	EP	3380876	A1 10/2018
CN	217639773	U	10/2022	EP	3385770	A1 10/2018
CN	217639774	U	10/2022	EP	3440484	A1 2/2019
CN	217639776	U	10/2022	EP	3504566	A2 7/2019
CN	217639777	U	10/2022	EP	3564747	A1 * 11/2019
CN	217639778	U	10/2022	EP	3631533	A1 4/2020
CN	217639903	U	10/2022	EP	3676973	A1 7/2020
CN	217639920	U	10/2022	EP	3743764	A1 12/2020
CN	115268058	A	11/2022	EP	3353578	B1 1/2021
CN	115327865	A	11/2022	EP	3799626	A1 4/2021
CN	115332917	A	11/2022	EP	3956702	A1 2/2022
CN	115343795	A	11/2022	EP	4004608	A1 6/2022
CN	115390176	A	11/2022	EP	4147311	A1 3/2023
CN	217809433	U	11/2022	EP	4268009	A1 11/2023
CN	217818613	U	11/2022	GB	2490895	A 11/2012
CN	217819022	U	11/2022	GB	2499869	B 3/2018
CN	217820828	U	11/2022	GB	2578049	A 4/2020
CN	217820829	U	11/2022	GB	2578233	A 4/2020
CN	217820831	U	11/2022	GB	2578236	A 4/2020
CN	217820834	U	11/2022	GB	2578236	B 11/2022
CN	217820838	U	11/2022	HK	40010538	7/2020
CN	217820839	U	11/2022	JP	2004302457	A 10/2004
CN	217820943	U	11/2022	JP	2005017408	A 1/2005
CN	217820944	U	11/2022	JP	2005274847	A 10/2005
CN	217820945	U	11/2022	JP	2008046428	A 2/2008
CN	217820971	U	11/2022	JP	2008299084	A 12/2008
CN	217821058	U	11/2022	JP	2010085977	A 4/2010
CN	217821068	U	11/2022	JP	2015502581	A 1/2015
				JP	2015092234	A 5/2015
				JP	105223689	A 1/2016
				JP	2016511936	A 4/2016
				JP	2017062373	A 3/2017

..... G03B 17/02

(56)

References Cited

FOREIGN PATENT DOCUMENTS

JP 2018536204 A 12/2018
 JP 2018537804 A 12/2018
 JP 2019516128 A 6/2019
 JP 2020522009 A 7/2020
 JP 2021511553 A 5/2021
 JP 6925358 B2 8/2021
 JP 2022542172 A 9/2022
 KR 20080099452 A 11/2008
 KR 20080103149 A 11/2008
 KR 20090002583 A 1/2009
 KR 20100027995 A 3/2010
 KR 101493928 B1 3/2015
 KR 20150113041 A 10/2015
 KR 20170015109 A 2/2017
 KR 20180083885 A 7/2018
 KR 20180121309 A 11/2018
 KR 20180124106 A 11/2018
 KR 101905444 B1 12/2018
 KR 20190038221 A 4/2019
 KR 102036640 B1 10/2019
 KR 1020200008630 A 1/2020
 KR 1020200108901 A 9/2020
 KR 20210088520 A 7/2021
 KR 10-2363805 B1 2/2022
 KR 1020220035971 A 3/2022
 SG 11201804346 P 11/2021
 SG 11201808772 W 11/2021
 SG 11202001717V 2/2023
 SG 11202006952X 2/2023
 TW 201017338 A 5/2010
 TW 201438242 A 10/2014
 TW 201908232 A 3/2019
 WO 2000043750 A2 7/2000
 WO 2007141788 A2 12/2007
 WO 2008019803 A1 2/2008
 WO 2008020899 A2 2/2008
 WO 2009067540 A1 5/2009
 WO 2009124181 A2 10/2009
 WO 2011106553 A2 9/2011
 WO 2011106553 A3 1/2012
 WO 2012122677 A1 9/2012
 WO 2012139634 A1 10/2012
 WO 2012144997 A1 10/2012
 WO 2012172366 A1 12/2012
 WO 2013033591 A1 3/2013
 WO 2014116500 A1 7/2014
 WO 2015021255 A1 2/2015
 WO 2015077926 A1 6/2015
 WO 2015112939 A1 7/2015
 WO 2015160412 A2 10/2015
 WO 2016049629 A1 3/2016
 WO 2016051325 A1 4/2016
 WO 2016086204 A1 6/2016
 WO 2016140720 A2 9/2016
 WO 2016140720 A3 10/2016
 WO 2016168173 A1 10/2016
 WO 2016178740 A2 11/2016
 WO 2016191142 A2 12/2016
 WO 2017005709 A1 1/2017
 WO 2017034995 A1 3/2017
 WO 2017040854 A1 3/2017
 WO 2017053309 A1 3/2017
 WO 2017079480 A1 5/2017
 WO 2017091738 A1 6/2017
 WO 2017176921 A1 10/2017
 WO 2017182771 A1 10/2017
 WO 2018063455 A1 4/2018
 WO 2018067246 A2 4/2018
 WO 2018063455 A9 5/2018
 WO 2018118984 A1 6/2018
 WO 2018134215 A1 7/2018
 WO 2018067246 A3 8/2018
 WO 2018142339 A1 8/2018
 WO 2018204856 A1 11/2018
 WO 2018218063 A1 11/2018

WO 2018219710 A1 12/2018
 WO 2018222944 A1 12/2018
 WO 2019015735 A1 1/2019
 WO 2019039241 A1 2/2019
 WO 2019043016 A1 3/2019
 WO 2019046827 A1 3/2019
 WO 2019057907 A1 3/2019
 WO 2019075335 A1 4/2019
 WO 2019101750 A2 5/2019
 WO 2019103762 A2 5/2019
 WO 2019108290 A1 6/2019
 WO 2019113106 A1 6/2019
 WO 2019116364 A1 6/2019
 WO 2019118646 A1 6/2019
 WO 2019119025 A1 6/2019
 WO 2019103762 A3 7/2019
 WO 2019136166 A1 7/2019
 WO 2019103762 A9 8/2019
 WO 2019147828 A1 8/2019
 WO 2019148200 A1 8/2019
 WO 2019164542 A1 8/2019
 WO 2019164849 A1 8/2019
 WO 2019173357 A1 9/2019
 WO 2019198568 A1 10/2019
 WO 2019203876 A2 10/2019
 WO 2019204667 A1 10/2019
 WO 2019206430 A1 10/2019
 WO 2020001938 A1 1/2020
 WO 2020010084 A1 1/2020
 WO 2020101568 A1 5/2020
 WO 2020139752 A1 7/2020
 WO 2020176227 A1 9/2020
 WO 2020214615 A1 10/2020
 WO 2020214617 A1 10/2020
 WO 2020248046 A1 12/2020
 WO 2021021671 A1 2/2021
 WO 2021130085 A1 7/2021
 WO 2021226544 A1 11/2021
 WO 2021230868 A1 11/2021
 WO 2022146929 A1 7/2022
 WO 2022150816 A1 7/2022
 WO 2023115037 A1 6/2023

OTHER PUBLICATIONS

Li et al., Metalens-Based Miniaturized Optical Systems, Mar. 31, 2019, Micromachines, 1-21 (Year: 2019).*

Arbabi et al., "Dielectric Metasurfaces for Complete Control of Phase and Polarization with Subwavelength Spatial Resolution and High Transmission", Nature Nanotechnology, Aug. 31, 2015, 27 pgs., doi:10.1038/nnano.2015.186.

Arbabi et al., "Miniature optical planar camera based on a wide-angle metasurface doublet corrected for monochromatic aberrations", Nature Communications, 2016, vol. 7, No. 13682, 9 pgs., doi:10.1038/ncomms13682.

Arbabi et al., "Subwavelength-thick lenses with high numerical apertures and large efficiency based on high-contrast transmitters", Nature Communications, May 5, 2015, vol. 6, pp. 7069, doi:10.1038/ncomms8069.

Azadegan et al., "A novel approach for miniaturization of slot antennas", IEEE Transactions on Antennas and Propagation, Mar. 2003, vol. 51, No. 3, pp. 421-429, doi:10.1109/TAP.2003.809853.

Blanchard et al., "Modeling nanoscale, V-shaped antennas for the design of optical phased arrays", Physical Review, Apr. 30, 2012, vol. B 85, pp. 155457-1-155457-11, DOI: 10.1103/physRevB.85.155457.

Buralli et al., "Optical Performance of Holographic Kinoforms", Applied Optics, Mar. 1, 1989, vol. 28, No. 5, pp. 976-983, doi:10.1364/AO.28.000976.

Byrnes et al., "Designing Large, High-Efficiency, High-Numerical-Aperture, Transmissive Meta-Lenses for Visible Light", Optics Express, Mar. 7, 2016, vol. 24, No. 5, pp. 5110-5124.

Campione et al., "Tailoring dielectric resonator geometries for directional scattering and Huygens' metasurface", Optics Express, Feb. 9, 2015, vol. 23, Issue 3, published online Jan. 28, 2015, pp. 2293-2304, arXiv:1410.2315, DOI: 10.1364/OE.23.002293.

(56)

References Cited

OTHER PUBLICATIONS

- Chen et al., "A broadband achromatic metalens for focusing and imaging in the visible", *Nature Nanotechnology*, Jan. 1, 2018, vol. 13, pp. 220-226, doi: 10.1038/s41565-017-0034-6.
- Chen et al., "A review of metasurfaces: physics and applications", *Reports on Progress in Physics*, Jun. 16, 2016, vol. 79, 076401, 40 pgs., doi: 10.1088/0034-4885/79/7/076401.
- Chen et al., "Dual-polarity plasmonic metalens for visible light", *Nature Communications*, Nov. 13, 2012, vol. 3, No. 1198, pp. 1-6, DOI 10.1038/ncomms2207.
- Chen et al., "High-Efficiency Broadband Meta-Hologram with Polarization-Controlled Dual Images", *Nano Letters*, 2014, vol. 14, No. 1, published online Dec. 13, 2013, pp. 225-230, <https://doi.org/10.1021/nl403811d>.
- Chen et al., "Immersion Meta-Lenses at Visible Wavelengths for Nanoscale Imaging", *Nano Letters*, Apr. 7, 2017, vol. 17, No. 5, pp. 3188-3194, doi: 10.1021/acs.nanolett.7b00717.
- Chen et al., "Phase and dispersion engineering of metalenses: broadband achromatic focusing and imaging in the visible", Nov. 26, 2017. Cornell University. [retrieved on Apr. 11, 2019]. Retrieved from the Internet: <URL:<https://arxiv.org/abs/1711.09343v1>>. entire document.
- Chou et al., "Imprint lithography with 25-nanometer resolution", *Science*, Apr. 5, 1996, vol. 272, Issue 5258, pp. 85-87.
- Dayal et al., "Polarization control of 0.85 μ m vertical-cavity surface-emitting lasers integrated with gold nanorod arrays", *Applied Physics Letters*, 2007, vol. 91, pp. 111107-1-111107-3, published online Sep. 12, 2007, DOI: 10.1063/1.2783281.
- Decker et al., "High-efficiency light-wave control with all-dielectric optical Huygens' metasurfaces", *Advanced Optical Materials*, arXiv:1405.5038, May 2014, pp. 813-820, doi:10.1002/adom.201400584.
- Devlin et al., "Arbitrary spin-to-orbital angular momentum conversion of light", *Science*, Nov. 17, 2017, vol. 358, pp. 896-901.
- Devlin et al., "Broadband high-efficiency dielectric metasurfaces for the visible spectrum", *Proceedings of the National Academy of Sciences of USA*, Sep. 20, 2016, vol. 113, No. 38, pp. 10473-10478, doi: 10.1073/pnas.1611740113.
- Devlin et al., "High Efficiency Dielectric Metasurfaces at Visible Wavelengths", Mar. 8, 2016 (Mar. 8, 2016), Retrieved from the Internet: <https://arxiv.org/ftp/arxiv/papers/1603/1603.02735.pdf>.
- Ding et al., "Gradient Metasurfaces: Fundamentals and Applications", arxiv.org, Cornell University Library, 2017, 83 pgs.
- Dong et al., "Zero-index photonic crystal as low-aberration optical lens (Conference Presentation)", *Proc. SPIE 9918, Metamaterials, Metadevices, and Metasystems*, Nov. 9, 2016, 991822, available at <https://doi.org/10.1117/12.2237137>, 1 pg.
- Evlyukhin et al., "Optical response features of Si-nanoparticle arrays", *Physical Review B*, 2010, vol. 82, 045404-1-045404-11, DOI: 10.1103/PhysRevB.82.045404.
- Fattal et al., "Flat dielectric grating reflectors with focusing abilities", *Nature Photonics*, May 2, 2010, vol. 4, No. 7, XP055162682, doi: 10.1038/nphoton.2010.116.
- Genevet et al., "Breakthroughs in Photonics 2013: Flat Optics: Wavefronts Control with Huygens' Interfaces", *IEEE Photonics Journal*, Apr. 1, 2014, vol. 6, No. 2, pp. 1-4, XP011546594, doi: 10.1109/jphot.2014.2308194.
- Genevet et al., "Recent advances in planar optics: from plasmonic to dielectric metasurfaces", *Optica*, Jan. 19, 2017, vol. 4, No. 1, doi: 10.1364/OPTICA.4.000139.
- Goldberg, "Genetic Algorithms in Search, Optimization, and Machine Learning", Addison-Wesley, 1989, 432 pgs., (presented in two parts).
- Groever et al., "Meta-Lens Doublet in the Visible Region", *Nano Letters*, Jun. 29, 2017, vol. 17, No. 8, pp. 4902-4907, doi: 10.1021/acs.nanolett.7b01888.
- Hartwig et al., "Challenges for Reducing the Size of Laser Activated Remote Phosphor Light Engines for DLP Projection", *Proceedings of SPIE, International Optical Design Conference*, Dec. 17, 2014, vol. 9293, pp. 929313-1 to 929313-6, doi: 10.1117/12.2073275, ISBN: 978-1-62841-730-2.
- Hidber et al., "Microcontact printing of Palladium colloids: micron-scale patterning by electroless deposition of copper", 1996, *Langmuir*, The ACS Journal of Surfaces and Colloids, vol. 12, pp. 1375-1380.
- Jin et al., "Waveforms for Optimal Sub-keV High-Order Harmonics with Synthesized Two- or Three-Colour Laser Fields", *Nature Communications*, May 30, 2014, vol. 5, No. 4003, pp. 1-6.
- Kats et al., "Giant birefringence in optical antenna arrays with widely tailorable optical anisotropy", *PNAS*, Jul. 31, 2012, vol. 109, No. 31, pp. 12364-12368, www.pnas.org/dgi/doi/10.1073/pnas.1210686109.
- Khorasaninejad et al., "Visible Wavelength Planar Metalenses Based on Titanium Dioxide", *IEEE Journal of Selected Topics in Quantum Electronics*, May/Jun. 2017, vol. 23, No. 3, pp. 43-58.
- Khorasaninejad et al., "Achromatic Metalens over 60 nm Bandwidth in the Visible and Metalens with Reverse Chromatic Dispersion", *Nano Letters*, Jan. 26, 2017, vol. 17, No. 3, pp. 1819-1824, doi: 10.1021/acs.nanolett.6b05137.
- Khorasaninejad et al., "Achromatic Metasurface Lens at Telecommunication Wavelengths", *Nano Letters*, Jul. 13, 2015, vol. 15, No. 8, pp. 5358-5362, doi: 10.1021/acs.nanolett.5b01727.
- Khorasaninejad et al., "Broadband and chiral binary dielectric meta-holograms", *Science Advances*, May 13, 2016, vol. 2, No. 5, 6 pgs.
- Khorasaninejad et al., "Broadband Multifunctional Efficient Meta-Gratings Based on Dielectric Waveguide Phase Shifters", *Nano Letters*, Sep. 15, 2015, vol. 15, No. 10, pp. 6709-6715, doi: 10.1021/acs.nanolett.5b02524.
- Khorasaninejad et al., "Metalenses at visible wavelengths: Diffraction-limited focusing and subwavelength resolution imaging", *Science*, Jun. 3, 2016, vol. 352, No. 6290, pp. 1190-1194, doi: 10.1126/science.aaf6644.
- Khorasaninejad et al., "Multispectral Chiral Imaging with a Metalens", *Nano Letters*, Jun. 7, 2016, vol. 16, pp. 4595-4600, doi: 10.1021/acs.nanolett.6b01897.
- Khorasaninejad et al., "Planar Lenses at Visible Wavelengths", *Arxiv*, May 7, 2016, 17 pages.
- Khorasaninejad et al., "Polarization-Insensitive Metalenses at Visible Wavelengths", *Nano Letters*, Oct. 24, 2016, vol. 16, No. 11, pp. 7229-7234, doi: 10.1021/acs.nanolett.6b03626.
- Khorasaninejad et al., "Silicon Nanofin Grating as a Miniature Chirality-Distinguishing Beam-Splitter", *Nature Communications*, 2014, vol. 5, No. 5386, p. No. 1-6, Published:—Nov. 12, 2014.
- Khorasaninejad et al., "Super-Dispersive Off-Axis Meta-Lenses for Compact High Resolution Spectroscopy", *Nano Letters*, Apr. 27, 2016, vol. 16, No. 6, pp. 3732-3737, doi: 10.1021/acs.nanolett.6b01097.
- Kildishev et al., "Planar Photonics with Metasurfaces", *Science*, Mar. 15, 2013, vol. 339, No. 6125, pp. 1232009-1-1232009-6.
- Kokkoris et al., "Nanoscale Roughness Effects at the Interface of Lithography and Plasma Etching: Modeling of Line-Edge-Roughness Transfer During Plasma Etching", *IEEE Transactions on Plasma Science*, Sep. 2009, vol. 37, No. 9, pp. 1705-1714.
- Kominami et al., "Dipole and Slot Elements and Arrays on Semi-Infinite Substrates", *IEEE Transactions on Antennas and Propagation*, Jun. 1985, vol. AP33, No. 6, pp. 600-607.
- Krasnok et al., "All-dielectric optical nanoantennas", *Optics Express*, Aug. 23, 2012, vol. 20, No. 18, pp. 20599-20604.
- Kress et al., "Applied Digital Optics from Micro-Optics to Nanophotonics", *Applied Digital Optics*, 2009, Wiley, 30 pgs.
- Lalanne et al., "Interaction between optical nano-objects at metallo-dielectric interfaces", *Nature Physics*, Aug. 2006, vol. 2, pp. 551-556, doi:10.1038/nphys364.
- Leveque et al., "Transient behavior of surface plasmon polaritons scattered at a subwavelength groove", *Physical Reviews B*, 76, Oct. 18, 2007, pp. 155418-1-155418-8, DOI: 10.1103/PhysRevB.76.155418.
- Lezec et al., "Beaming Light from a Subwavelength Aperture", *Science Express*, Aug. 2, 2002, vol. 297, pp. 820-822.

(56)

References Cited

OTHER PUBLICATIONS

- Li et al., "Achromatic Flat Optical Components via Compensation between Structure and Material Dispersions", *Scientific Reports*, 2016, vol. 6, No. 19885, 7 pgs., DOI:10.1038/srep19885.
- Li et al., "Flat metasurfaces to focus electromagnetic waves in reflection geometry", *Optics Letters*, 2012, vol. 37, No. 23, pp. 4940-4942.
- Lin et al., "Dielectric gradient metasurface optical elements", *Science*, Jul. 18, 2014, vol. 345, Issue 6194, pp. 298-302, DOI: 10.1126/science.1253213.
- Liu et al., "Realization of polarization evolution on higher-order Poincare sphere with metasurface", *Applied Physics Letters*, 2014, vol. 104, pp. 191110-1-191101-4, <http://dx.doi.org/10.1063/1.4878409>.
- Lo et al., "New Architecture for Space Telescopes Uses Fresnel Lenses", *SPIE Newsroom*, Aug. 9, 2006, 2 pgs., doi: 10.1117/2.1200608.0333.
- Lu et al., "Planar high-numerical-aperture low-loss focusing reflectors and lenses using subwavelength high contrast gratings", *Optics Express*, Jun. 7, 2010, vol. 18, No. 12, pp. 12606-12614, doi: 10.1364/OE.18.012606.
- Luk et al., "Dielectric Resonator Antennas", *Research Studies Press LTD, Hertfordshire*, 2003, 404 pgs. (presented in two parts).
- Mao et al., "Nanopatterning Using a Simple Bi-Layer Lift-Off Process for the Fabrication of a Photonic Crystal Nanostructure", *Nanotechnology*, Feb. 1, 2013, vol. 24, No. 8, 6 pgs., doi:10.1088/0957-4484/24/8/085302.
- Mao et al., "Surface Patterning of Nonscattering Phosphors for Light Extraction", *Optics Letters*, Aug. 1, 2013, vol. 38, No. 15, pp. 2796-2799, doi: 10.1364/OL.38.002796.
- Martin-Moreno, "Theory of highly directional emission from a single sub-wavelength aperture surrounded by surface corrugations", *Physical Review Letters*, Apr. 25, 2003, vol. 90, No. 16, 167401, pp. 167401-1-167401-4, published online Apr. 23, 2003, doi:10.1103/PhysRevLett.9.167401.
- McLeod, "Thin-Film Optical Filters", *Adam Hilger*, 1986, 667 pgs. (presented in four parts).
- Miyazaki et al., "Ultraviolet-Nanoimprinted Packaged Metasurface Thermal Emitters for Infrared CO₂ Sensing", *Science and Technology of Advanced Materials*, Published May 20, 2015, vol. 16, No. 3, pp. 5, doi: 10.1088/1468-6996/16/3/035005.
- Monticone et al., "Full Control of Nanoscale Optical Transmission with a Composite Metascreen", *Physical Review Letters*, May 17, 2013, vol. 110, pp. 203903-1-203903, DOI: 10.1103/PhysRevLett.110.203903.
- Mueller et al., "Metasurface Polarization Optics: Independent Phase Control of Arbitrary Orthogonal States of Polarization", *Physical Review Letters*, Mar. 17, 2017, vol. 118, 113901, 5 pgs.
- Ni et al., "Broadband Light Bending with Plasmonic Nanoantennas", *Science*, Jan. 27, 2012, vol. 335, Issue 6067, 3 pgs., published online Dec. 22, 2011, DOI: 10.1126/science.1214686.
- Ni et al., "Ultra-thin, planar, Babinet-inverted plasmonic metalenses", *Light Science & Applications*, 2013, vol. 2, e72, pp. 1-6, published online Apr. 26, 2013, doi:10.1038/lsa.2013.28.
- Okaya et al., "The Dielectric Microwave Resonator", *Proceedings of the IRE*, Oct. 1962, vol. 50, Issue 10, pp. 2081-2092, DOI: 10.1109/JRPROC.1962.288245.
- Pacheco-Peña et al., "Epsilon-near-zero metalenses operating in the visible", *Optics & Laser Technology*, Jan. 19, 2016, 80, pp. 162-168.
- Peinado et al., "Optimization and performance criteria of a Stokes polarimeter based on two variable retarders", *Optics Express*, Apr. 12, 2010, vol. 18, No. 8, pp. 9815-9530.
- Petosa et al., "An Overview of Tuning Techniques for Frequency-Agile Antennas", *IEEE Antennas and Propagation Magazine*, Oct. 2012, vol. 52, pp. 5, pp. 271-296.
- Pfeiffer et al., "Metamaterial Huygens' Surface: Tailoring Wave Fronts with Reflectionless Sheets", *Physical Review Letters*, May 10, 2013, vol. 110, pp. 197401-1-197401-5, DOI: 10.1103/PhysRevLett.110.197401.
- Pors et al., "Broadband Focusing Flat Mirrors Based on Plasmonic Gradient Metasurfaces", *Nano Letters*, Jan. 23, 2013, vol. 13, No. 2, pp. 829-834, <https://doi.org/10.1021/nl304761m>.
- Reichelt et al., "Capabilities of diffractive optical elements for real-time holographic displays", *Proceedings of SPIE*, Feb. 2008, vol. 6912, pp. 69120-69130, <http://dx.doi.org/10.1117/12.762887>.
- Rubin et al., "Polarization State Generation and Measurement with a Single Metasurface", *Optics Express*, Aug. 20, 2018, vol. 26, Issue No. 17, pp. 21455-21478.
- Saeidi et al., "Wideband plasmonic focusing metasurfaces", *Applied Physics Letters*, Aug. 2014, vol. 105, pp. 053107-1-053107-4, <http://dx.doi.org/10.1063/1.4892560>.
- Sales et al., "Diffractive-Refractive Behavior of Kinoform Lenses", *Applied Optics*, Jan. 1, 1997, vol. 36, pp. 253-257, No. 1, doi: 10.1364/AO.36.000253.
- Sancho-Parramon et al., "Optical characterization of HfO₂ by spectroscopic ellipsometry: dispersion models and direct data inversion", *Thin Solid Films*, 2008, vol. 516, pp. 7990-7995, available online Apr. 10, 2008, doi:10.1016/j.tsf.2008.04.007.
- She et al., "Large area metalenses: design, characterization, and mass manufacturing", *Optics Express*, Jan. 22, 2018, vol. 26, No. 2, pp. 1573-1585, doi: 10.1364/OE.26.001573.
- Sun et al., "High-Efficiency Broadband anomalous Reflection by Gradient Meta-Surfaces", *Nano Letters*, 2012, vol. 12, No. 12, pp. 6223-6229, dx.doi.org/10.1021/nl3032668.
- Vo et al., "Sub-wavelength grating lenses with a twist", *IEEE Photonics Technology Letters*, Jul. 1, 2014, vol. 26, No. 13, pp. 1375-1378, DOI: 10.1109/LPT.2014.2325947.
- Voelz, "Computational Fourier Optics: A Matlab® Tutorial", (SPIE Press, 2011), 23 pgs.
- Walther et al., "Spatial and Spectral Light Shaping with Metamaterials", *Advanced Materials*, 2012, vol. 24, pp. 6300-6304, doi: 10.1002/adma.201202540.
- Wang et al., "Generation of steep phase anisotropy with zero-backscattering by arrays of coupled dielectric nano-resonators", *Applied Physics Letters*, 2014, vol. 105, pp. 121112-1-121112-5, published online Sep. 25, 2014, <https://doi.org/10.1063/1.4896631>.
- Wu et al., "Spectrally selective chiral silicon metasurfaces based on infrared Fano resonances", *Nature Communications*, May 27, 2014, vol. 5, No. 3892, pp. 1-9, doi:10.1038/ncomms4892.
- Yang et al., "Design of ultrathin plasmonic quarter-wave plate based on period coupling", *Optics Letters*, 2013, vol. 38, No. 5, pp. 679-681, <https://doi.org/10.1364/OL.38.000679>.
- Yao et al., "Wide Wavelength Tuning of Optical Antennas on Graphene with Nanosecond Response Time", *Nano Letters*, 2014, First Published Dec. 3, 2013, vol. 14, No. 1, pp. 214-219, doi: 10.1021/nl403751p.
- Yu et al., "A Broadband, Background-Free Quarter-Wave Plate Based on Plasmonic Metasurfaces", *Nano Letters*, Nov. 3, 2012, vol. 12, No. 12, pp. 6328-6333, dx.doi.org/10.1021/nl303445u.
- Yu et al., "Flat optics with designer metasurfaces", *Nature Materials*, Feb. 2014, vol. 13, pp. 139-150, published online Jan. 23, 2014, DOI:10.1038/NMAT3839.
- Yu et al., "Flat optics: Controlling wavefronts with optical antenna metasurfaces", *IEEE Journal of Selected Topics*, 2013, vol. 19, No. 3, 23 pgs.
- Yu et al., "High-Transmission Dielectric Metasurface with 2 Phase Control at Visible Wavelengths", *Laser & Photonics Reviews*, Jun. 26, 2015, No. 4, pp. 412-418.
- Yu et al., "Light Propagation with Phase Discontinuities: Generalized Laws of Reflection and Refraction", *Science*, Oct. 21, 2011, vol. 334, No. 6054, pp. 333-337, doi: 10.1126/science.1210713.
- Yu et al., "Optical Metasurfaces and Prospect of their Applications Including Fiber Optics", *Journal of Lightwave Technology*, 2015, vol. 33, No. 12, pp. 2344-2358, XP011584804.
- Yu et al., "Quantum cascade lasers with integrated plasmonic antenna-array collimators", *Optics Express*, Nov. 24, 2008, vol. 16, No. 24, pp. 19447-19461, published online Nov. 10, 2008.
- Yu et al., "Small divergence edge-emitting semiconductor lasers with two-dimensional plasmonic collimators", *Applied Physics Letters*, 2008, vol. 93, pp. 181101-1-181101-3, doi: 10.1063/1.3009599.

(56)

References Cited**OTHER PUBLICATIONS**

- Yu et al., "Small-divergence semiconductor lasers by plasmonic collimation", *Nature Photonics*, vol. 2, pp. 564-570, Year 2008.
- Zhan et al., "Low-contrast dielectric metasurface optics", *ACS Photonics* 2016, 3, 209-214. DOI: 10.1021/acsp Photonics.5b00660.
- Zhao et al., "Mie resonance-based dielectric metamaterials", *Materials Today*, Dec. 2009, vol. 12, No. 12, pp. 60-69.
- Zhao et al., "Recent Advances on Optical Metasurfaces", *Journal of Optics*, Institute of Physics Publishing, 2014, vol. 16, Issue 12, 14 pgs., doi:10.1088/2040489781/16/12/123001.
- Zhao et al., "Twisted Optical metamaterials or planarized ultrathin broadband circular polarizers", *Nature Communications*, 2012, vol. 3, No. 870, pp. 1-7, DOI: 10.1038/ncomms1877.
- Zhou et al., "Characteristic Analysis of Compact Spectrometer Based on Off-Axis Meta-Lens", *Applied Sciences*, 2018, vol. 8, vol. 321, doi:10.3390/app8030321, 11 pgs.
- Zhou et al., "Plasmonic holographic imaging with V-shaped nanoantenna array", *Optics Express*, Feb. 25, 2013, vol. 21, No. 4, pp. 4348-4354, published online Feb. 12, 2013.
- Zhu et al., "Ultra-compact visible chiral spectrometer with meta-lenses", *APL Photonics*, Feb. 7, 2017, vol. 2, pp. 036103-1-036103-12, 13 pgs., doi: 10.1063/1.4974259.
- Zou et al., "Dielectric resonator nanoantennas at visible frequencies", *Optics Express*, Jan. 14, 2013, vol. 21, No. 1, pp. 1344-1352, published online Jan. 11, 2013.
- International Preliminary Report on Patentability for International Application PCT/US2020/043600, Report issued Feb. 1, 2022, Mailed on Feb. 10, 2022, 08 Pgs.
- Cumme et al., "From Regular Periodic Micro-Lens Arrays to Randomized Continuous Phase Profiles", *Adv. Opt. Techn.*, 2015, vol. 4, No. 1, pp. 47-61.
- Orazbayev et al., "Tunable Beam Steering Enabled by Graphene Metamaterials", *Optics Express* 8848, 2016, vol. 24, No. 8, 14 pgs., DOI:10.1364/OE.24.008848.
- Sayyah et al., "Two-dimensional pseudo-random optical phased array based on tandem optical injection locking of vertical cavity surface emitting lasers", *Optics Express* 19405, Jul. 27, 2015, vol. 23, No. 15, 12 pgs., DOI:10.1364/OE.23.019405.
- Seurin et al., "High-Efficiency VCSEL Arrays for Illumination and Sensing in Consumer Applications", *Proc. SPIE 9766, Vertical-Cavity Surface-Emitting Lasers XX*, 97660D, Mar. 4, 2016, pp. 97660D-1-97660D-9, doi:10.1117/12.2213295.
- Silvestri et al., "Robust design procedure for dielectric resonator metasurface lens array", *Optics Express*, Dec. 12, 2016, vol. 24, No. 25, 29154, 17 pgs.
- Voelkel et al., "Laser Beam Homogenizing: Limitations and Constraints", *DPIE, Europe, Optical Systems Design*, 2008, 12 pgs.
- Zhou et al., "Progress on Vertical-Cavity Surface-Emitting Laser Arrays for Infrared Illumination Applications", *Proc. SPIE 9001, Vertical-Cavity Surface-Emitting Lasers XVIII*, 90010E, Feb. 27, 2014, 11 pgs., doi: 10.1117/12.2040429.
- Search Report for Chinese Patent Application No. 201680077924.9, mailed on Aug. 30, 2019, 10 Pages.
- Search Report for Chinese Patent Application No. 201780031669.9, mailed on Mar. 4, 2020, 29 Pages.
- Extended European Search Report for European Application 17858861.2, Report Completed Mar. 13, 2020, Mailed Mar. 23, 2020, 09 Pgs.
- Extended European Search Report for European Application No. 17779772.7, Search completed Oct. 15, 2019, Mailed Oct. 25, 2019, 10 Pgs.
- Extended European Search Report for European Application No. 18852460.7, Search completed Mar. 25, 2021, Mailed Apr. 6, 2021, 13 Pgs.
- Extended European Search Report for European Application No. 16869282.0, Search completed Nov. 8, 2019, Mailed Nov. 20, 2019, 15 Pgs.
- Extended Search Report for European Application No. 18805669.1, Search completed Feb. 9, 2021, Mailed Feb. 18, 2021, 13 Pgs.
- International Preliminary Report on Patentability for International Application PCT/US2018/049276, Report issued on Mar. 3, 2020, Mailed on Mar. 12, 2020, 8 Pgs.
- International Preliminary Report on Patentability for International Application PCT/US2019/040302, Report issued Jan. 5, 2021, Mailed Jan. 14, 2021, 5 Pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2018/038357, Report issued Dec. 24, 2019, Mailed Jan. 2, 2020, 6 Pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2012/053434, Report issued Mar. 4, 2014, Mailed Mar. 13, 2014, 6 Pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2015/064930, Report issued Jun. 13, 2017, Mailed Jun. 22, 2017, 8 Pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2016/063617, Report issued May 29, 2018, Mailed Jun. 7, 2018, 6 Pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2017/026206, Report issued Oct. 9, 2018, Mailed Oct. 18, 2018, 8 Pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2018/031204, Report issued Nov. 5, 2019, Mailed Nov. 14, 2019, 8 Pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2018/035502, Report issued Dec. 3, 2019, Mailed Dec. 12, 2019, 7 Pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2018/046947, Issued Feb. 18, 2020, mailed on Feb. 27, 2020, 6 Pgs.
- International Preliminary Report on Patentability for International Application PCT/US2008/084068, Report issued on May 25, 2010, 5 Pgs.
- International Preliminary Report on Patentability for International Application PCT/US2016/052685, Report issued Mar. 27, 2018, Mailed Apr. 5, 2018, 8 Pgs.
- International Preliminary Report on Patentability for International Application PCT/US2017/036897, Report issued Dec. 11, 2018, Mailed Dec. 20, 2018, 8 Pgs.
- International Preliminary Report on Patentability for International Application PCT/US2017/048469, Report issued Feb. 26, 2019, Mailed Mar. 7, 2019, 5 Pgs.
- International Preliminary Report on Patentability for International Application PCT/US2018/034460, Report issued Nov. 26, 2019, Mailed Dec. 5, 2019, 6 Pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2012/053434, Search completed Oct. 17, 2012, Mailed Dec. 17, 2012, 7 Pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2019/014975, Search completed Jun. 17, 2019, Mailed Jul. 8, 2019, 10 pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2019/018615, Search completed Apr. 12, 2019, Mailed May 6, 2019, 12 Pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2019/040302, completed Aug. 29, 2019, Mailed Oct. 17, 2019, 6 Pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2020/043600, Search completed Sep. 29, 2020, Mailed Nov. 24, 2020, 11 Pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2008/084068, Completed Jan. 13, 2009, Mailed Feb. 2, 2009, 6 pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2015/064930, Search completed Sep. 9, 2016, Mailed Sep. 20, 2016, 11 Pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2016/052685, Search completed Nov. 30, 2016, Mailed Dec. 9, 2016, 12 Pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2016/063617, Search completed Jan. 19, 2017, Mailed Mar. 31, 2017, 9 Pgs.

(56)

References Cited**OTHER PUBLICATIONS**

International Search Report and Written Opinion for International Application No. PCT/US2017/026206, Search completed Jun. 10, 2017, Mailed Jun. 28, 2017, 11 Pgs.

International Search Report and Written Opinion for International Application No. PCT/US2017/036897, Search completed Jan. 31, 2018, Mailed Feb. 21, 2018, 9 Pgs.

International Search Report and Written Opinion for International Application No. PCT/US2017/048469, Search completed Apr. 20, 2018, Mailed May 4, 2018, 9 Pgs.

International Search Report and Written Opinion for International Application No. PCT/US2018/031204, Search completed Jun. 29, 2018, Mailed Jul. 23, 2018, 14 Pgs.

International Search Report and Written Opinion for International Application No. PCT/US2018/034460, Search completed Jul. 29, 2018, Mailed Aug. 24, 2018, 10 Pgs.

International Search Report and Written Opinion for International Application No. PCT/US2018/035502, Search completed Jul. 31, 2018, Mailed Aug. 24, 2018, 13 Pgs.

International Search Report and Written Opinion for International Application No. PCT/US2018/038357, Search completed Apr. 9, 2019, Mailed May 13, 2019, 12 Pgs.

International Search Report and Written Opinion for International Application No. PCT/US2018/046947, Search completed Oct. 14, 2019, Mailed Oct. 25, 2019, 10 Pgs.

International Search Report and Written Opinion for International Application No. PCT/US2018/049276, Search completed Oct. 26, 2018, Mailed Jan. 15, 2019, 12 Pgs.

Office Action for Chinese Patent Application No. 201680077924.9, dated Aug. 30, 2019, 10 pgs.

Search Report and Written Opinion for International Application No. 11201808772W, Search completed Jan. 20, 2020, Mailed Jan. 28, 2020, 12 Pgs.

Supplementary Partial European Search Report for European Application No. 16869282.0, Search completed Jun. 19, 2019, Mailed Jul. 2, 2019, 12 Pgs.

"Materials for High and Low Refractive Index Coatings", Sigma-Aldrich tech. www.sigmaaldrich.com/materials-science/organic-electronics/ri-coatings.html (3 pages).

"These Tiny, Incredible 'Metalenses' are the Next Giant Leap in Optics", PetaPixel, Jun. 3, 2016, 21 pgs.

Aieta et al., "Aberration-Free Ultrathin Flat Lenses and Axicons at Telecom Wavelengths Based on Plasmonic Metasurfaces", Nano Lett., Web publication date Aug. 15, 2012, vol. 12, No. 9, pp. 4932-4936.

Aieta et al., "Aberrations of flat lenses and aplanatic metasurfaces", Optics Express, Dec. 16, 2013, vol. 21, No. 25, pp. 31530-31539, doi: 10.1364/oe.21.031530.

Aieta et al., "Multiwavelength achromatic metasurfaces by dispersive phase compensation", Science Express Reports, Feb. 19, 2015, 8 pgs., doi: 10.1126/science.aaa2494.

Aieta et al., "Out-of-Plane Reflection and Refraction of Light by Anisotropic Optical Antenna Metasurfaces with Phase Discontinuities", Nano Letters, Feb. 15, 2012, vol. 12, No. 3, pp. 1702-1706, doi: 10.1021/nl300204s.

Azzam, R. M. A. "Stokes-Vector and Mueller-Matrix Polarimetry [Invited]", Journal of the Optical Society of America A, vol. 33, No. 7, Jul. 2016, 1396-1408.

Azzam et al., "Accurate Calibration of the Four-Detector Photopolarimeter with Imperfect Polarizing Optical Elements", Journal of the Optical Society of America A, vol. 6, No. 10, Oct. 1989, pp. 1513-1521.

Azzam et al., "Photopolarimeter Based on Planar Grating Diffraction", Journal of the Optical Society of America A, vol. 10, No. 6, Jun. 1993, pp. 1190-1196.

Bao et al., "Toward the Capacity Limit of 2D Planar Jones Matrix with a Single-Layer Metasurface", Science Advances, vol. 7, Jun. 18, 2021, pp. 1-6, doi: 10.1126/sciadv.abh0365.

Berry et al., "Measurement of the Stokes Parameters of Light", Applied Optics, vol. 16, No. 12, Dec. 1977, pp. 3200-3205.

Bomzon et al., "Real-Time Analysis of Partially Polarized Light with a Space-Variant Subwavelength Dielectric Grating", Optics Letters, vol. 27, No. 3, Feb. 1, 2002, pp. 188-190.

Bomzon et al., "Spatial Fourier-Transform Polarimetry Using Space-Variant Subwavelength Metal-Stripe Polarizers", Optics Letters, vol. 26, No. 21, Nov. 1, 2001, pp. 1711-1713.

Capaldo et al., "Nano-Fabrication and Characterization of Silicon Meta-Surfaces Provided with Pancharatnam-Berry effect", Optical Materials Express, Mar. 1, 2019, vol. 9, No. 3, pp. 1015-1032.

Chen et al., "Broadband Achromatic Metasurface-Refractive Optics", Nano Letters, vol. 18, Nov. 13, 2018, pp. 7801-7808.

Chen et al., "Engineering the Phase Front of Light with Phase-Change Material Based Planar Lenses", Scientific Reports, vol. 5, No. 8660, Mar. 2, 2015, pp. 1-7.

Chen et al., "Integrated Plasmonic Metasurfaces for Spectropolarimetry", Nanotechnology, vol. 27, Apr. 26, 2016, pp. 1-7, doi:10.1088/0957-4484/27/22/224002.

Chen et al., "Supplementary Information of Engineering the Phase Front of Light with Phase-Change Material Based Planar Lenses", Scientific Reports, 2015, 4 pgs.

Chipman et al., "Polarized Light and Optical Systems", CRC Press, 2019, 106 pgs.

Chou et al., "Subwavelength Amorphous Silicon Transmission Gratings and Applications in Polarizers and Waveplates", Applied Physics Letters, vol. 67, No. 6, Aug. 7, 1995, pp. 742-744.

Cincotti, "Polarization Gratings: Design and Applications", IEEE Journal of Quantum Electronics, vol. 39, 2003, pp. 1645-1652.

Cloude, "Conditions for the Physical Realisability of Matrix Operators in Polarimetry", Proceedings of SPIE, vol. 1166, 1989, pp. 177-185, doi: 10.1117/12.962889.

Cui et al., "Sixteen-Beam Grating-Based Division-of-Amplitude Photopolarimeter", Optics Letters, vol. 21, No. 1, Jan. 1, 1996, pp. 89-91.

Davis et al., "Diffraction Gratings Generating Orders with Selective States of Polarization", Optics Express, vol. 24, No. 2, 2016, pp. 907-917, doi: 10.1364/OE.24.000907.

Davis et al., "Polarization Beam Splitters Using Polarization Diffraction Gratings", Optics Letters, vol. 26, No. 9, May 1, 2001, pp. 587-589.

Deng et al., "Diatomic Metasurface for Vectorial Holography", Nano Letters, Mar. 28, 2018, pp. A-H.

Deschapms et al., "The Polder Mission: Instrument Characteristics and Scientific Objectives", IEEE Transactions on Geoscience and Remote Sensing, vol. 32, No. 3, May 1994, pp. 598-615.

Ding et al., "Beam-Size-Invariant Spectropolarimeters Using Gap-Plasmon Metasurfaces", ACS Photonics, vol. 4, Feb. 28, 2017, pp. 943-949.

Ding et al., "Versatile Polarization Generation and Manipulation Using Dielectric Metasurfaces", Laser & Photonics Review, vol. 14, Sep. 23, 2020, pp. 2000116-1-2000116-7.

Fienup, J. R., "Phase Retrieval Algorithms: A Comparison", Applied Optics, vol. 21, No. 15, Aug. 1, 1982, pp. 2758-2769.

Gerchberg et al., "A Practical Algorithm for the Determination of Phase from Image and Diffraction Plane Pictures", Optik, vol. 35, No. 2, 1972, pp. 1-6.

Gori, Franco, "Measuring Stokes Parameters by Means of a Polarization Grating", Optics Letters, vol. 24, No. 9, May 1, 1999, pp. 584-586.

Gutiérrez-Vega, Julio C., "Optical Phase of Inhomogeneous Jones Matrices: Retardance and Ortho-Transmission States", Optics Letters, vol. 45, No. 7, Apr. 1, 2020, pp. 1639-1642, doi: 10.1364/OL.387644.

Hasman et al., "Chapter 4: Space-Variant Polarization Manipulation", Progress in Optics, vol. 47, 2005, pp. 215-289, doi: 10.1016/S0079-6638(05)47004-3.

Herrera-Fernandez et al., "Double Diffractive Optical Element System for Near-Field Shaping", Applied Optics, vol. 50, No. 23, Aug. 10, 2011, pp. 4587-4593.

Horie et al., "Reflective Optical Phase Modulator Based on High-Contrast Grating Mirrors", Optical Society of America, IEEE, 2014, 2 pgs.

Hsiao et al., "Fundamentals and Applications of Metasurfaces", Small Methods, vol. 1, Mar. 24, 2017, pp. 1600064-1-1600064-20.

(56)

References Cited

OTHER PUBLICATIONS

- Jang et al., "Wavefront Shaping with Disorder-Engineered Metasurfaces", *Nature Photonics*, 2018, 8 pgs.
- Juan et al., "Arbitrary Polarization Transformation Based on Two-Dimensional Metallic Rectangular Gratings", *Acta Optica Sinica*, Dec. 31, 2011, vol. 31, No. 12, 1224001-1-1224001-5. Doi:10.3788/AOS201131.1224001.
- Karagodsky et al., "Monolithically Integrated Multi-Wavelength VCSEL Arrays Using High-Contrast Gratings", *Optics Express*, vol. 18, No. 2, Jan. 18, 2010, pp. 694-699, doi: <https://doi.org/10.1364/OE.18.000694>.
- Keller, "Instrumentation for Astrophysical Spectropolarimetry" *Astrophysical Spectropolarimetry*, 2001, pp. 303-354.
- Lee et al., "Giant Nonlinear Response from Plasmonic Metasurfaces Coupled to Intersubband Transitions", *Nature*, vol. 511, Jul. 3, 2014, pp. 65-69.
- Li et al., "All-Silicon Nanorod-Based Dammann Gratings", *Optics Letters*, vol. 40, No. 18, Sep. 15, 2015, pp. 4285-4288.
- Li et al., "Broadband Diodelike Asymmetric Transmission of Linearly Polarized Light in Ultrathin Hybrid Metamaterial", *Applied Physics Letters*, vol. 105, Nov. 19, 2014, pp. 201103-1-201103-5, doi: 10.1063/1.4902162.
- Li et al., "Dispersion Controlling Meta-Lens at Visible Frequency", *Optics Express*, vol. 25, No. 18, Sep. 4, 2017, pp. 21419-21427.
- Liu et al., "Single-Pixel Computational Ghost Imaging with Helicity-Dependent Metasurface Hologram", *Science Advances*, vol. 3, No. E1701477, Sep. 8, 2017, pp. 1-6.
- Liu et al., "SSD: Single Shot Multibox Detector", *European Conference on Computer Vision*, Springer, 2016, pp. 21-37, doi: 10.1007/978-3-319-46448-0_2.
- Lizana et al., "Arbitrary State of Polarization with Customized Degree of Polarization Generator", *Optics Letters*, vol. 40, No. 16, Aug. 15, 2015, pp. 3790-3793.
- Lohmann, A. W., "Reconstruction of Vectorial Wavefronts", *Applied Optics*, vol. 4, No. 12, 1965, pp. 1667-1668.
- Lu et al., "Homogeneous and Inhomogeneous Jones Matrices", *Journal of the Optical Society of America A*, 1994, vol. 11, No. 2, pp. 766-773.
- Lu et al., "Interpretation of Mueller Matrices Based on Polar Decomposition", *Journal of the Optical Society of America A*, vol. 13, No. 5, May 1996, pp. 1106-1113.
- Mackus et al., "The Use of Atomic Layer Deposition in Advanced Nanopatterning", *Nanoscale*, vol. 6, Jul. 25, 2014, 10941-10960.
- Maguid et al., "Multifunctional Interleaved Geometric-Phase Dielectric Metasurfaces", *Light: Science & Applications*, vol. 6, No. E17027, Aug. 11, 2017, pp. 1-7, doi: 10.1038/lsa.2017.27.
- Maguid et al., "Photonic Spin-Controlled Multifunctional Shared-Aperture Antenna Array", *Science*, vol. 352, No. 6290, Apr. 21, 2016, pp. 1202-1206.
- Meng et al., "A Novel Nanofabrication Technique of Silicon-Based Nanostructures", *Nanoscale Research Letters* vol. 11, No. 504, pp. 1-9, doi:10.1186/s11671-016-1702-4.
- Menzel et al., "Advanced Jones Calculus for the Classification of Periodic Metamaterials", *Physical Review A*, vol. 82, Nov. 15, 2010, pp. 053811-1-053811-9, doi: 10.1103/PhysRevA.82.053811.
- Mirsalehi, Mir M., "Optical Information Processing", *Encyclopedia of Physical Science and Technology*, 3rd edition, 2001, pp. 335-340.
- Moreno et al., "Jones Matrix Treatment for Optical Fourier Processors with Structured Polarization", *Optics Express*, vol. 19, No. 5, Feb. 28, 2011, pp. 4583-4594.
- Moreno et al., "Jones matrix treatment for polarization Fourier optics", *Journal of Modern Optics*, Mar. 26, 2004, Vol. 51, No. 14, pp. 2031-2038.
- Mueller et al., "Ultracompact Metasurface In-Line Polarimeter", *Optica*, vol. 3, No. 1, Jan. 8, 2016, pp. 42-47.
- Nordin et al., "Micropolarizer Array for Infrared Imaging Polarimetry", *Journal of the Optical Society of America A*, vol. 16, No. 5, May 1999, pp. 1168-1174.
- Novikova et al., "Polarimetric Imaging for Cancer Diagnosis and Staging", *Optics and Photonics News*, Oct. 2012, 8 pgs.
- Oh et al., "Achromatic Diffraction from Polarization Gratings with High Efficiency", *Optic Letters*, vol. 33, No. 20, Oct. 15, 2008, pp. 2287-2289.
- Otten et al., "The Vector Apodizing Phase Plate Coronagraph: Prototyping, Characterization and Outlook", *Advances in Optical and Mechanical Technologies for Telescopes and Instrumentation*, *Proceedings of SPIE*, vol. 9151, 2014, pp. 91511R-1-91511R-10.
- Pfeiffer et al., "Cascaded Metasurfaces for Complete Phase and Polarization Control", *Applied Physics Letters*, vol. 102, Jun. 11, 2013, pp. 231116-1-231116-4, doi: 10.1063/1.4810873.
- Pierangelo et al., "Polarimetric Imaging of Uterine Cervix: A Case Study", *Optics Express*, vol. 21, No. 12, Jun. 17, 2013, pp. 14120-14130.
- Pors et al., "Plasmonic Metagratings for Simultaneous Determination of Stokes Parameters", *Optica*, vol. 2, No. 8, Aug. 2015, pp. 716-723.
- Pors et al., "Plasmonic Metagratings for Simultaneous Determination of Stokes Parameters", *Arxiv:1609.04691v1 [physics.optics]*, Sep. 15, 2016, 21 pgs., doi: <http://dx.doi.org/10.1364/OPTICA.2.000716>.
- Pors et al., "Waveguide Metacouplers for In-Plane Polarimetry", *Physical Review Applied*, vol. 5, 2016, pp. 064015-1-064015-9.
- Ramos et al., "Error Propagation in Polarimetric Demodulation", *Applied Optics*, vol. 47, No. 14, May 10, 2008, pp. 2541-2549.
- Redding et al., "Full-Field Interferometric Confocal Microscopy Using a VCSEL Array", *Optics Letters*, vol. 39, No. 15, Aug. 1, 2014, 11 pgs.
- Romero et al., "Theory of Optimal Beam Splitting by Phase Gratings. II. Square and Hexagonal Gratings", *Journal of the Optical Society of America A*, vol. 24, No. 8, Aug. 2007, pp. 2296-2312.
- Rosales-Guzman et al., "How to Shape Light with Spatial Light Modulators", *SPIE*, 2017, 58 pgs.
- Rubin et al., "Matrix Fourier Optics Enables a Compact Full-Stokes Polarization Camera", *Science*, vol. 365, No. 43, Jul. 5, 2019, pp. 1-10, doi: 10.1126/science.aax1839.
- Sabatke et al., "Optimization of Retardance for a Complete Stokes Polarimeter", *Optics Letters*, vol. 25, No. 11, Jun. 1, 2000, pp. 802-804.
- Schulz et al., "Quantifying the Impact of Proximity Error Correction on Plasmonic Metasurfaces", *Optical Materials Express*, vol. 5, No. 12, Dec. 1, 2015, pp. 2798-2803, doi: 10.1364/OME.5.002798.
- Sell et al., "Periodic Dielectric Metasurfaces with High-Efficiency, Multiwavelength Functionalities", *Advanced Optical Materials*, 2017, 7 pgs., doi: 10.1002/adom.201700645.
- Shi et al., "Continuous Angle-Tunable Birefringence with Freeform Metasurfaces for Arbitrary Polarization Conversion", *Science Advances*, vol. 6, No. eaba3367, Jun. 3, 2020, pp. 1-7, doi: 10.1126/sciadv.aba3367.
- Shim et al., "Hard-Tip, Soft-Spring Lithography", *Nature*, vol. 469, Jan. 27, 2011, pp. 516-521.
- Snik et al., "An Overview of Polarimetric Sensing Techniques and Technology with Applications to Different Research Fields", *Proceedings of SPIE*, 2014, vol. 9099, pp. 90990B-1-90990B-20.
- Sokolov, A. L., "Polarization of Spherical Waves", *Optics and Spectroscopy*, vol. 92, No. 6, 2002, pp. 936-942.
- Song et al., "Vividly-Colored Silicon Metasurface Based on Collective Electric and Magnetic Resonances", *IEEE*, Jan. 11, 2016, 2 pgs.
- Sreelal et al., "Jones Matrix Microscopy from a Single-Shot Intensity Measurement", *Optics Letters*, Dec. 15, 2017, vol. 42, Issue 24, pp. 5194-5197.
- Stenflo, J. O., "Chapter 13: Instrumentation for Solar Polarimetry", *Solar Magnetic Fields: Polarized Radiation Diagnostics*, Springer Netherlands, Dordrecht, 1994, pp. 312-350.
- Tervo et al., "Paraxial-Domain Diffractive Elements with 100% Efficiency Based on Polarization Gratings", *Optics Letters*, vol. 25, No. 11, Jun. 1, 2000, pp. 785-786.
- Cai et al., "Structured light field 3D imaging", *Opt. Express* 2016, vol. 24, Issue 18, pp. 20324-20334.
- Chen et al., "Distance measurement based on light field geometry and ray tracing", *Optics Express*, 2017, vol. 25, issue 1, pp. 59-76.

(56)

References Cited**OTHER PUBLICATIONS**

- Cofre et al., "Quantitative performance of a polarization diffraction grating polarimeter encoded onto two liquid-crystal-on-silicon displays", *Optics & Laser Technology*, vol. 96, 2017, pp. 219-226.
- Damask, "Polarization Optics in Telecommunications", Springer, 2005, 535 pgs. (Presented in 2 Parts).
- Huang et al., "Polarimetric target depth sensing in ambient illumination based on polarization-coded structured light", *Applied Optics* 2017, vol. 56, Issue 27, pp. 7741-7748.
- Huang et al., "Target enhanced 3D reconstruction based on polarization-coded structured light", *Optics Express*, 2017, vol. 25, Issue 2, pp. 1173-1184.
- Nikolova et al., "Polarization Holography", Bulgarian Academy of Sciences, Sofia, P. S. Ramanujam, Technical University of Denmark, Roskilde Cambridge University, Aug. 2009.
- Peinado et al., "Polarization imaging with enhanced spatial resolution", *Optics Communications*, vol. 338, Mar. 2015, pp. 95-100.
- Extended European Search Report for European Application No. 20847649.9, Search completed Jul. 20, 2023, Mailed Aug. 1, 2023, 11 Pgs.
- Lim et al., "Self-Mixing Imaging Sensor Using a Monolithic VCSEL Array with Parallel Readout", *Optics Express*, vol. 17, No. 7, Mar. 30, 2009, pp. 5517-5525.
- Martin-Regalado et al., "Polarization Properties of Vertical-Cavity Surface-Emitting Lasers", *IEEE Journal of Quantum Electronics*, vol. 33, No. 5, May 1997, pp. 765-783, doi: 10.1109/3.572151.
- Su et al., "Designing LED Array for Uniform Illumination Distribution by Simulated Annealing Algorithm", *Optics Express*, vol. 20, No. S6, Nov. 5, 2012, pp. A843-A855.
- Xu et al., "Metasurface Quantum-Cascade Laser with Electrically Switchable Polarization", *Optica*, vol. 4, No. 4, Apr. 2017, pp. 468-475.
- Birch et al., "3D Imaging with Structured Illumination for Advanced Security Applications", United States: N. p., 2015. Web. doi:10.2172/1221516. <https://www.osti.gov/biblio/1221516>. 64 pgs.
- Roy et al., "Sub-wavelength focusing meta-lens", *Optics Express*, 2013, vol. 21, pp. 7577-7582.
- Wen et al., "Metasurface for characterization of the polarization state of light", *Optics Express*, 2015, vol. 23, No. 8, pp. 10272-10281.
- Extended European Search Report for European Application No. 19744012.6, Search completed Sep. 3, 2021, Mailed Dec. 16, 2021, 10 pgs.
- Extended European Search Report for European Application No. 19830958.5, Search completed Feb. 17, 2022, Mailed Feb. 25, 2022, 8 pgs.
- Extended European Search Report for European Application No. 20790964.9, Search completed Nov. 22, 2022, Mailed Dec. 2, 2022, 10 pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2020/028157, Report issued Sep. 28, 2021, Mailed Oct. 28, 2021, 7 pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2020/028159, Report issued Sep. 28, 2021, Mailed on Oct. 28, 2021, 7 pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2022/070043, Report issued Jul. 4, 2023, Mailed on Jul. 20, 2023, 10 pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2020/028157, Search completed Jun. 16, 2020, Mailed Sep. 4, 2020, 9 pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2020/028159, Search completed Jun. 15, 2020, Mailed Aug. 11, 2020, 7 pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2021/031423, Search completed Jul. 15, 2021, Mailed Aug. 16, 2021, 7 pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2021/065231, Search completed Apr. 19, 2022, Mailed May 13, 2022, 14 pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2022/038059, Search completed Sep. 27, 2022, Mailed Oct. 27, 2022, 17 pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2022/070043, Search completed May 5, 2022, Mailed Jun. 9, 2022, 16 pgs.
- International Search Report and Written Opinion for International Application No. PCT/US2022/081868, Search completed Mar. 23, 2023, Mailed Apr. 4, 2023, 13 pgs.
- Partial European Search Report for European Application No. 19744012.6, Search completed Sep. 3, 2021, Mailed Sep. 15, 2021, 12 pgs.
- "Elliptical Polarization", Wikipedia, XP055893535, Jan. 11, 2022, retrieved from the Internet URL: <https://en.wikipedia.org/wiki/Elliptical_polarization>, retrieved on Feb. 18, 2022, 4 pgs.
- Andreou et al., "Polarization Imaging: Principles and Integrated Polarimeters", *IEEE Sensors Journal*, vol. 2, No. 6, 2002, pp. 566-576.
- Arbabi et al., "Dielectric Metasurfaces for Complete Control of Phase and Polarization with Subwavelength Spatial Resolution and High Transmission", *Nat. Nanotechnol.*, vol. 10, Nov. 2014, pp. 937-943.
- Arbabi et al., "Efficient Dielectric Metasurface Collimating Lenses for Mid-Infrared Quantum Cascade Lasers", *Optics Express*, vol. 23, No. 26, Dec. 28, 2015, pp. 33310-33317, doi: 10.1364/OE.23.033310.
- Arbabi et al., "Full-Stokes Imaging Polarimetry Using Dielectric Metasurfaces", *ACS Photonics*, Jul. 16, 2018, vol. 5, No. 8, pp. 3132-3140.
- Arbabi et al., "Supplementary Figures of Miniature Optical Planar Camera Based on a Wide-Angle Metasurface Doublet Corrected for Monochromatic Aberrations", *Nature Communications*, 2016, vol. 7, Article No. 13682, 9 pgs.
- Arbabi et al., "Vectorial Holograms with a Dielectric Metasurface: Ultimate Polarization Pattern Generation", *ACS Photonics*, vol. 6, 2019, pp. 2712-2718, doi: 10.1021/acsphotonics.9b00678.
- Arrizon et al., "Pixelated Phase Computer Holograms for the Accurate Encoding of Scalar Complex Fields", *Journal of the Optical Society of America A*, vol. 24, No. 11, 2007, pp. 3500-3507.
- Azzam, R. M. A., "Arrangement of Four Photodetectors for Measuring the State of Polarization of Light", *Optics Letters*, vol. 10, No. 7, Jul. 1985, pp. 309-311.
- Azzam, R. M. A., "Division-of-Amplitude Photopolarimeter (DOAP) for the Simultaneous Measurement of All Four Stokes Parameters of Light", *Optica Acta*, vol. 29, No. 5, 1982, pp. 685-689.
- Todorov et al., "Polarization Holography. I: A New High-Efficiency Organic Material with Reversible Photoinduced Birefringence", *Applied Optics*, vol. 23, No. 23, Dec. 1, 1984, pp. 4309-4312.
- Todorov et al., "Spectrophotopolarimeter: Fast Simultaneous Real-Time Measurement of Light Parameters", *Optics Letters*, vol. 17, Mar. 1, 1992, pp. 358-359.
- Trebino et al., "The Autocorrelation, the Spectrum, and Phase Retrieval", *Frequency-Resolved Optical Gating: The Measurement of Ultrashort Laser Pulses*, (Springer, 2000), pp. 61-99.
- Tyo, Scott J., "Noise Equalization in Stokes Parameter Images Obtained by Use of Variable-Retardance Polarimeters", *Optics Letters*, vol. 25, No. 16, Aug. 15, 2000, pp. 1198-1200.
- Tyo et al., "Review of Passive Imaging Polarimetry for Remote Sensing Applications", *Applied Optics*, vol. 45, Aug. 1, 2006, pp. 5453-5469.
- Wang et al., "Broadband Achromatic Optical Metasurface Devices", *Nature Communications*, vol. 8, No. 187, Aug. 4, 2017, pp. 1-9, doi: 10.1038/s41467-017-00166-7.
- Wang et al., "Information Authentication Using an Optical Dielectric Metasurface", *Journal of Physics D: Applied Physics*, Institute of Physics Publishing, Bristol, vol. 50, No. 36, Aug. 17, 2017, pp. 1-5.
- Wei et al., "Design of Ultracompact Polarimeters based on Dielectric Metasurfaces", *Optics Letters*, vol. 42, No. 8, Apr. 11, 2017, pp. 1580-1583, doi: <https://doi.org/10.1364/OL.42.001580>.
- Wiktorowicz et al., "Toward the Detection of Exoplanet Transits with Polarimetry", *The Astrophysical Journal*, vol. 795, No. 12, Nov. 1, 2014, 6 pgs., doi: 10.1088/0004-637X/795/1/12.

(56)

References Cited

OTHER PUBLICATIONS

Wolf, "Introduction to the Theory of Coherence and Polarization of Light", Cambridge University Press, 2007, 235 pgs.

Xu et al., "Metasurface External Cavity Laser", *Applied Physics Letters*, vol. 107, No. 221105, 2015, pp. 221105-1-221105-5, doi: 10.1063/1.4936887.

Yang et al., "Generalized Hartmann-Shack Array of Dielectric Metalens sub-arrays for Polarimetric Beam Profiling", *Nature Communications*, Nov. 2, 2018, vol. 9, No. 907, 7 pp. 1-7, doi: 10.1038/s41467-018-07056-6.

Yariv et al., *Photonics: Optical Electronics in Modern Communications*, 6th edition (Oxford University Press, 2006), 849 pgs.

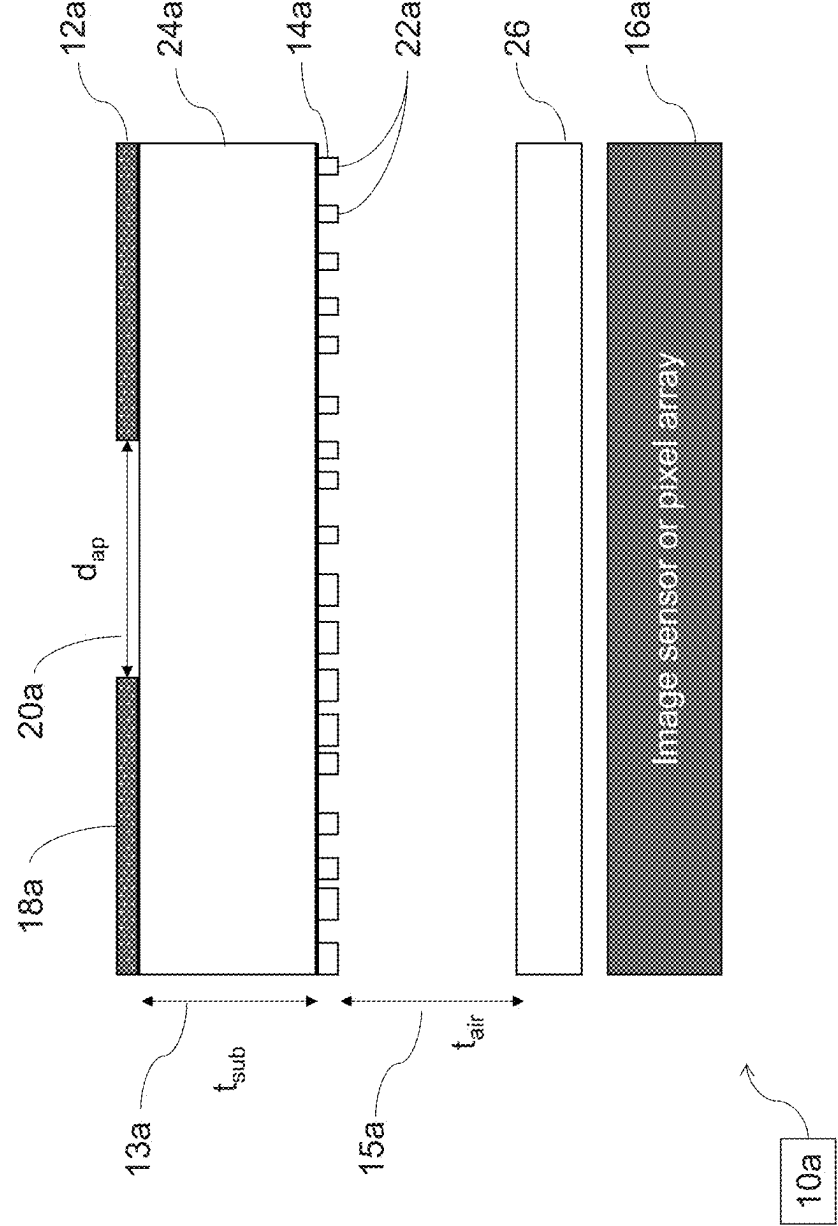
Yun et al., "Skew Aberration: a Form of Polarization Aberration", *Optics Letters*, vol. 36, No. 20, pp. 4062-4064, doi: 10.1364/OL.36.004062.

Zhang et al., "High Efficiency all-Dielectric Pixelated Metasurface for Near-Infrared Full-Stokes Polarization Detection", *Photonics Research*, vol. 9, No. 4, Apr. 2021, pp. 583-589, doi: <https://doi.org/10.1364/PRJ.415342>.

Zhao et al., "Multichannel Vectorial Holographic Display and Encryption", *Light Science & Applications*, vol. 7, No. 95, 2018, 9 pgs., doi: 10.1038/s41377-018-0091-0.

Zhao et al., "Tailoring the Dispersion of Plasmonic Nanorods to Realize Broadband Optical Meta-Waveplates", *Nano Letter*, vol. 13, Feb. 5, 2013, pp. 1086-1091, doi: [dx.doi.org/10.1021/nl304392b](https://doi.org/10.1021/nl304392b).

* cited by examiner



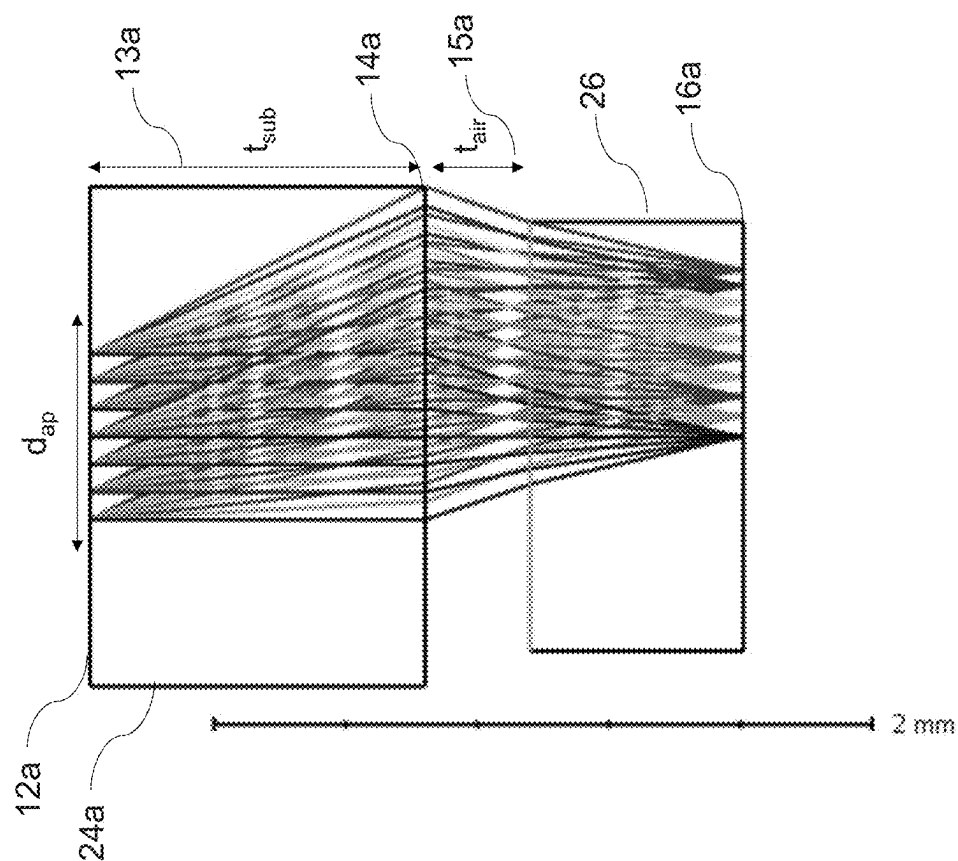
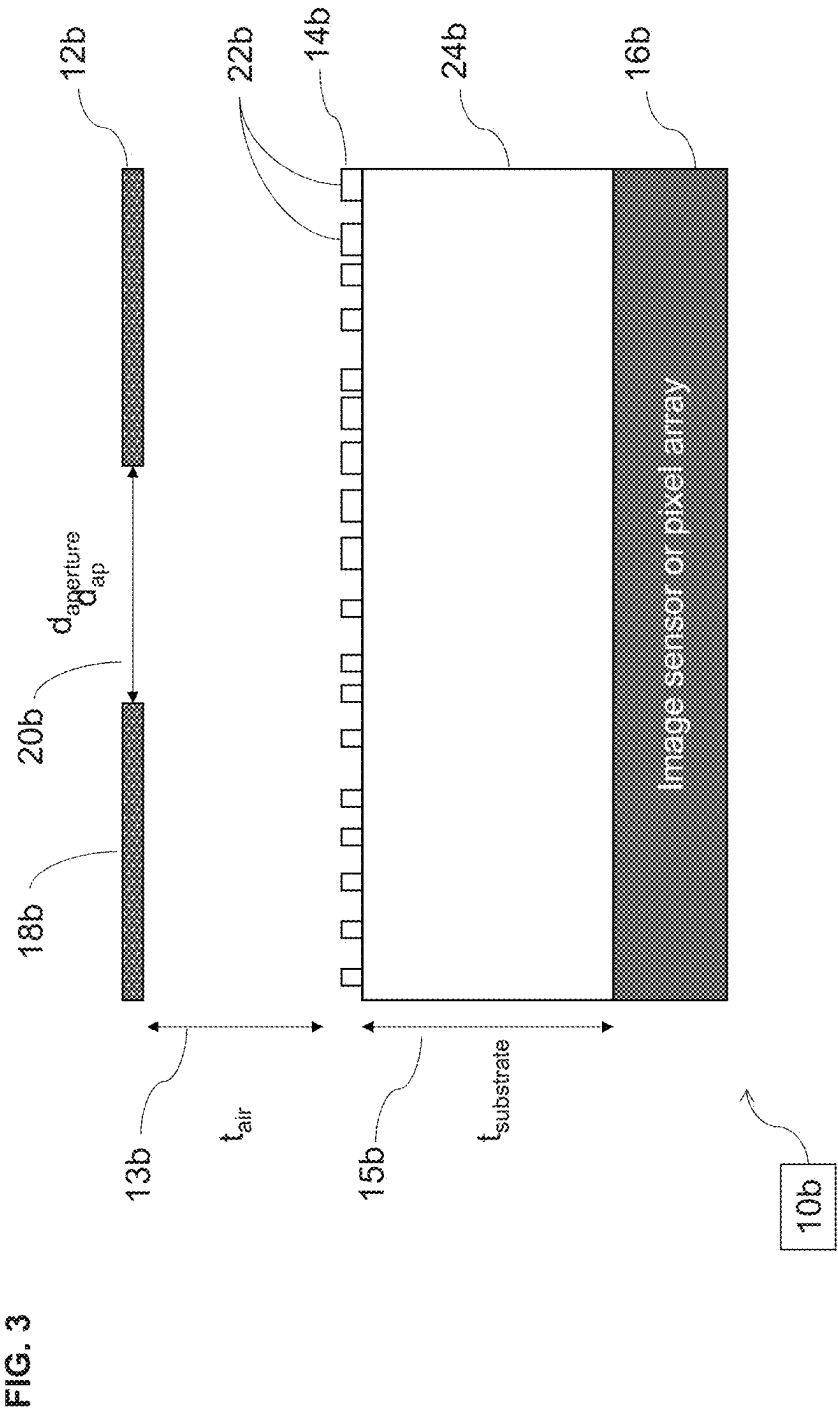


FIG. 2



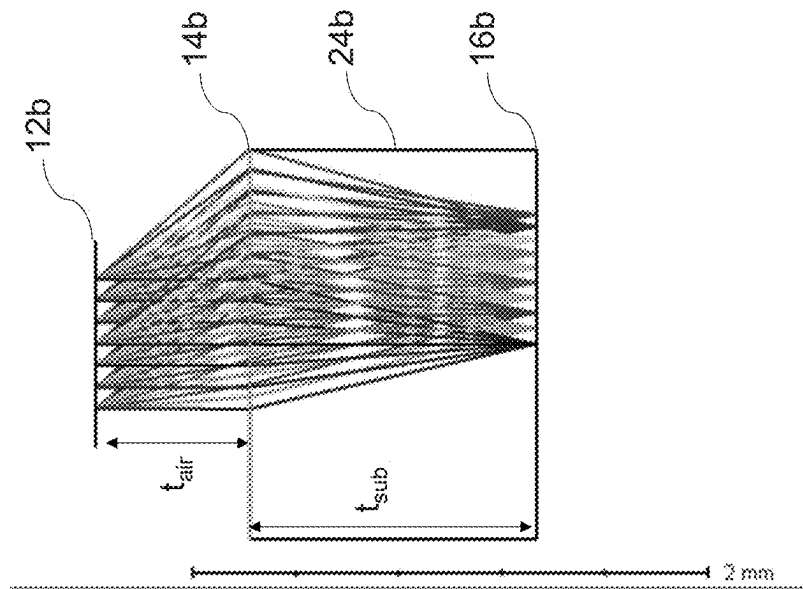


FIG. 4

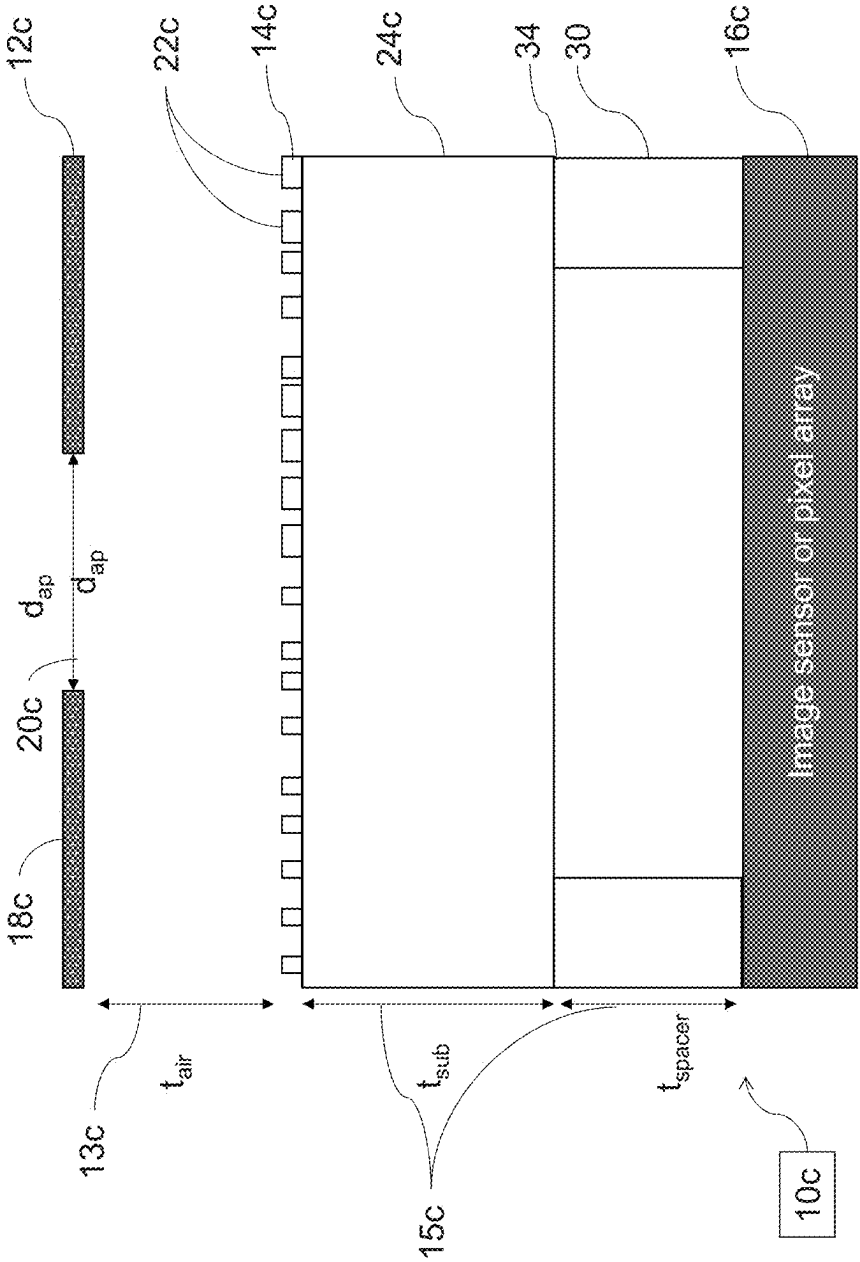
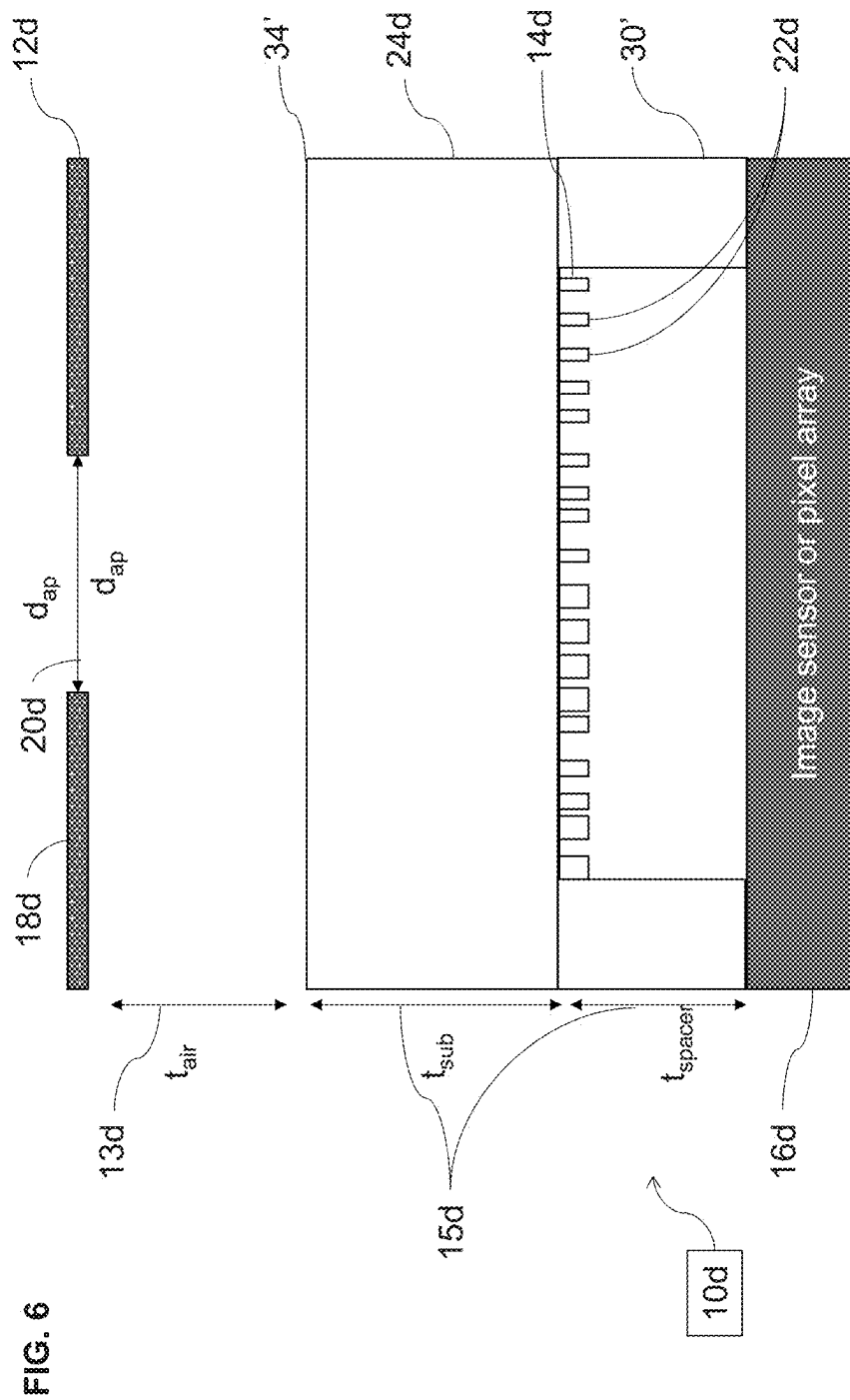


FIG. 5



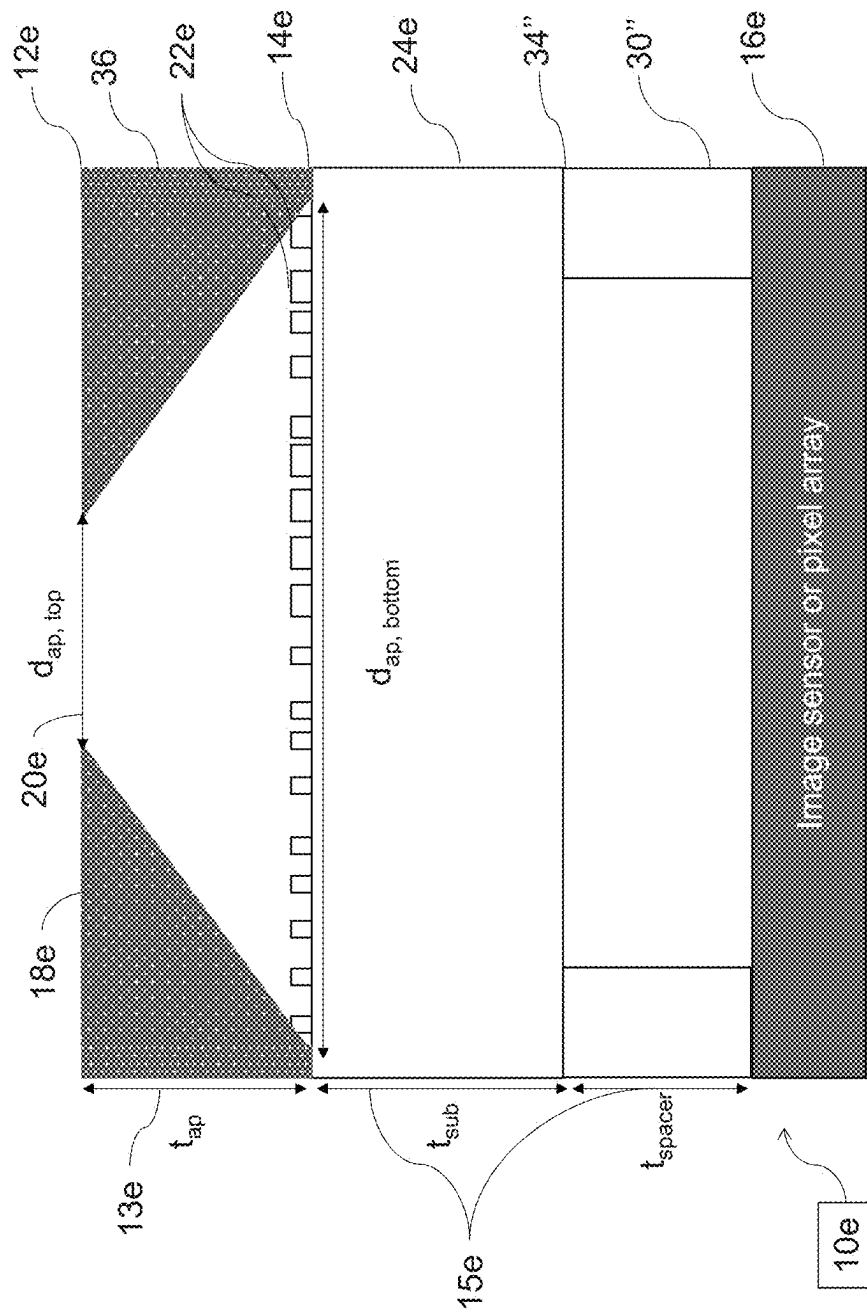


FIG. 7

FIG. 8B

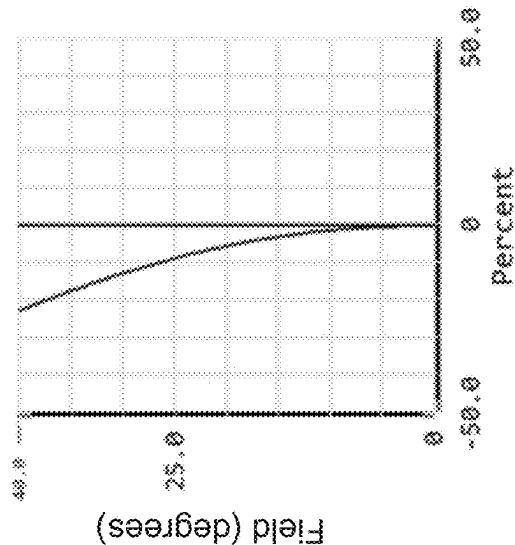


FIG. 8A

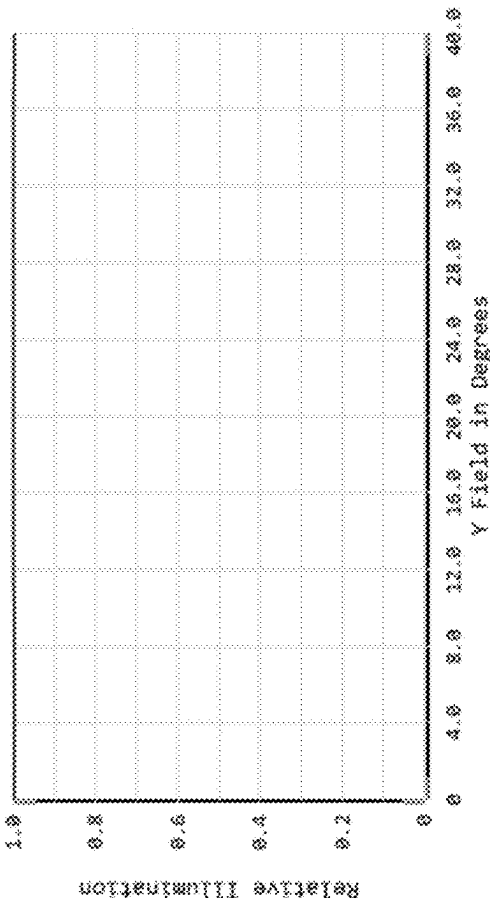
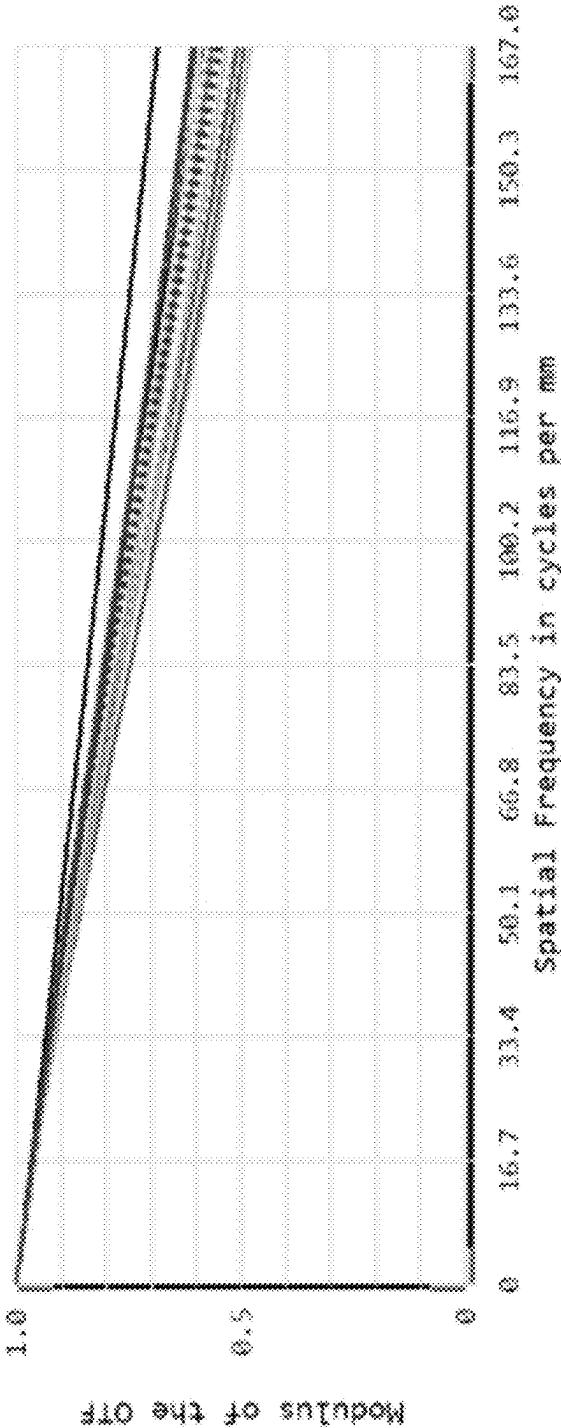


FIG. 8C



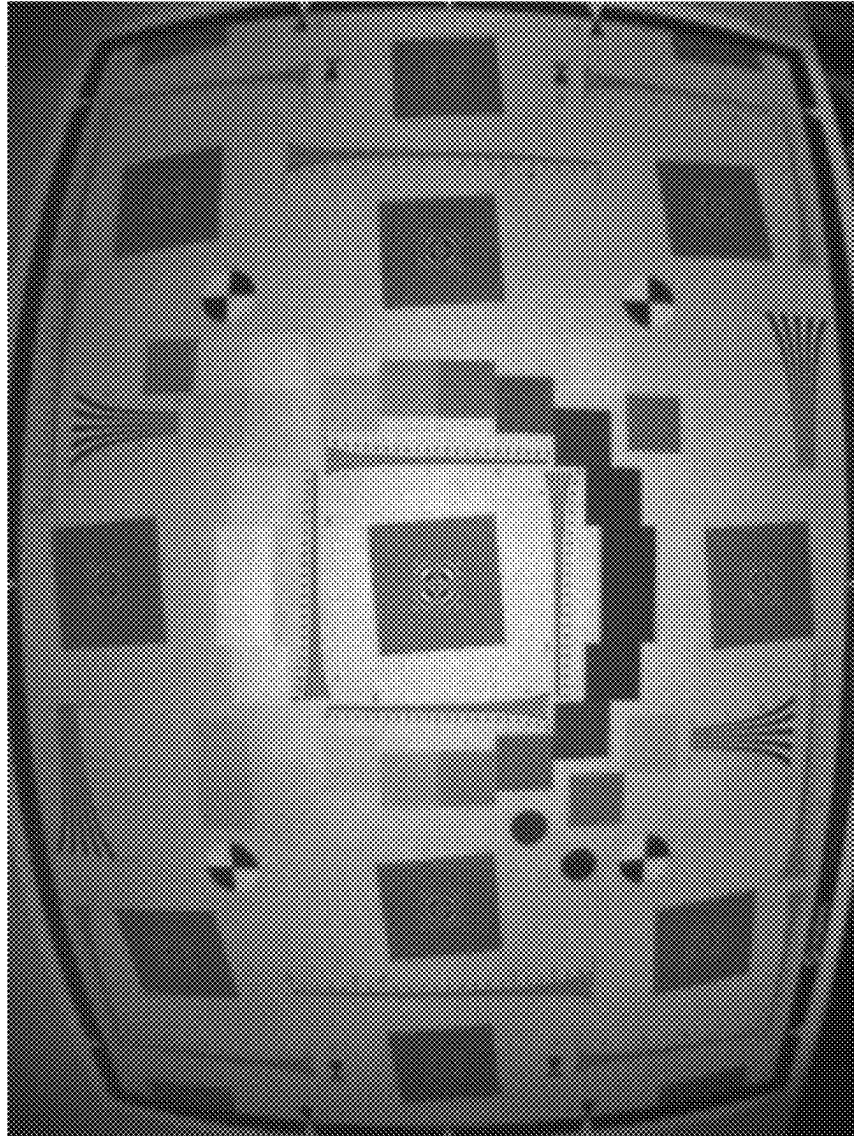


FIG. 8D

FIG. 9

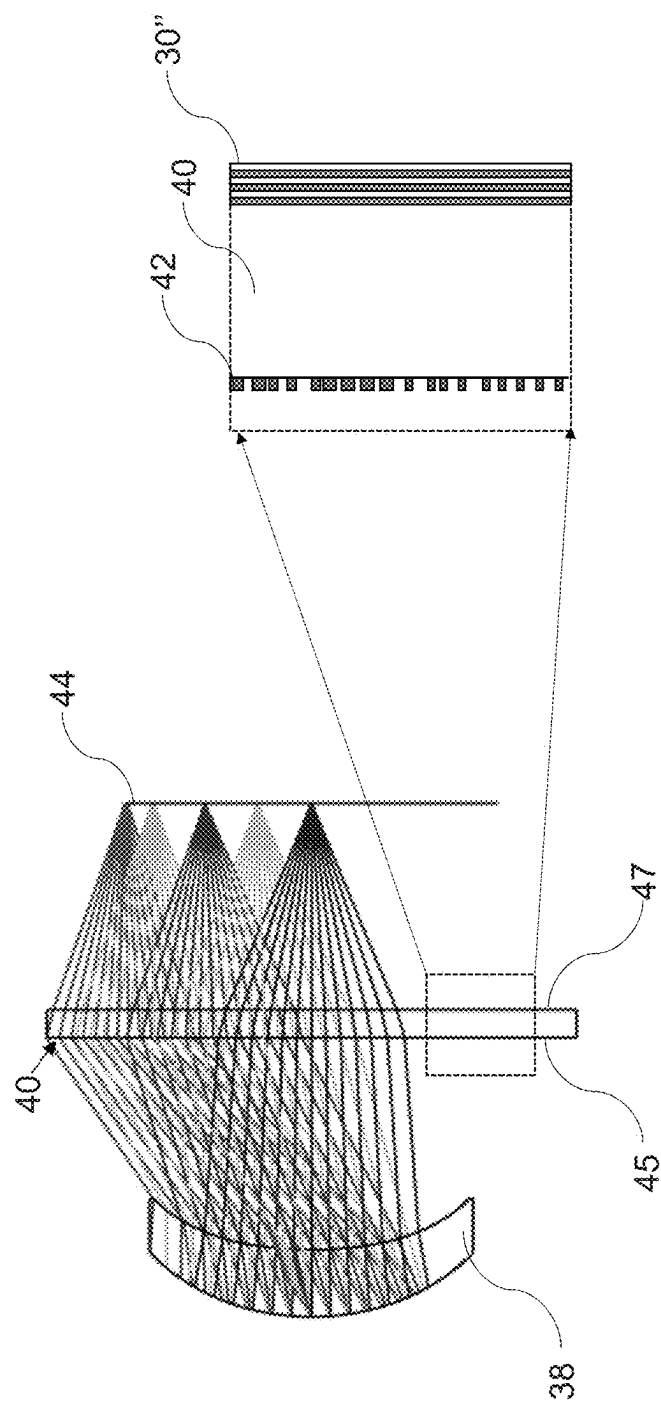


FIG. 10B

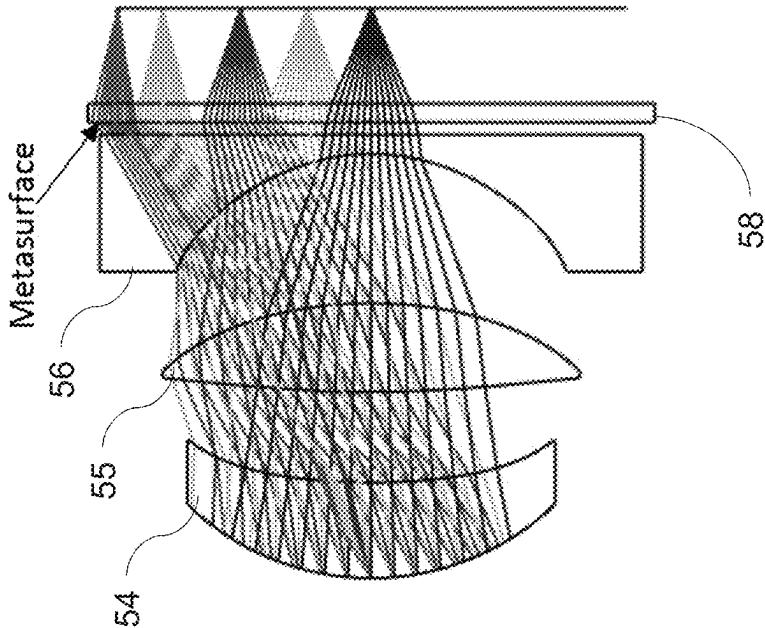


FIG. 10A

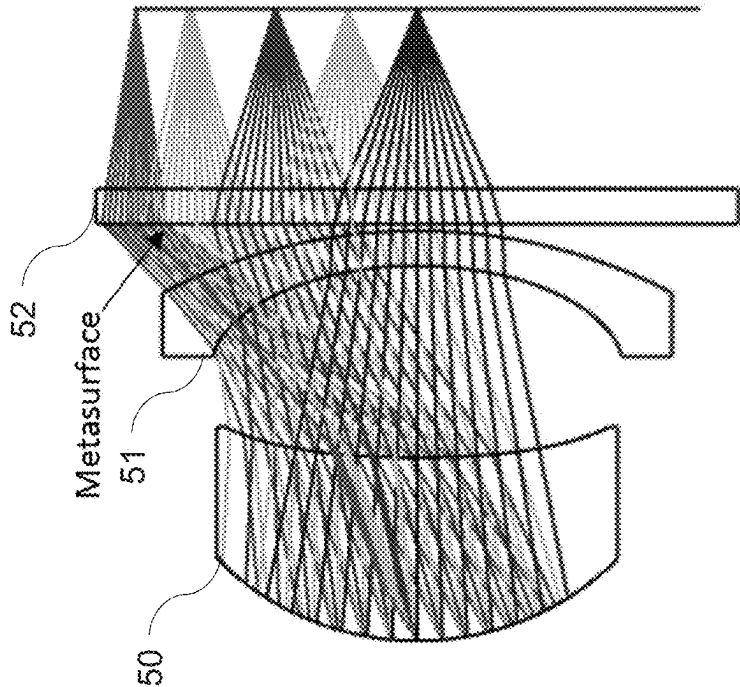


FIG. 11B

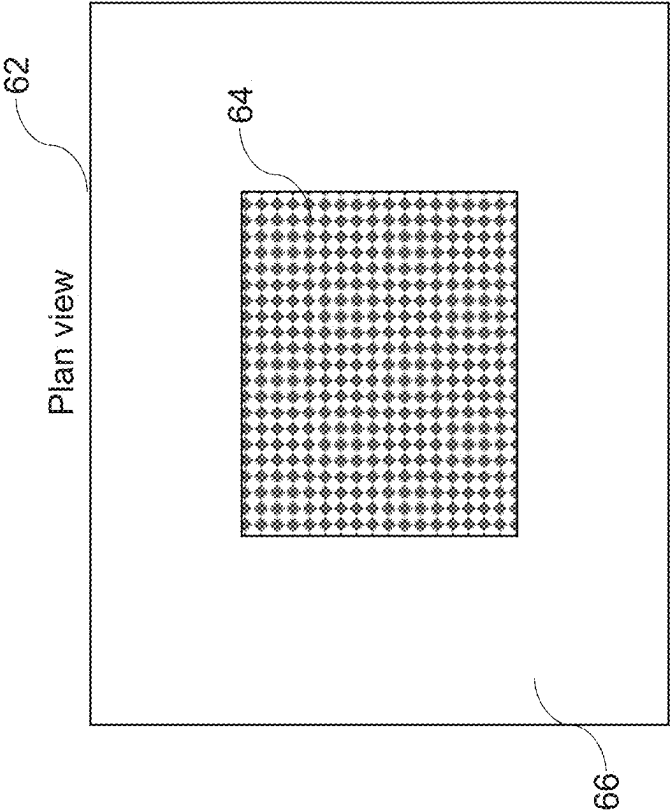


FIG. 11A

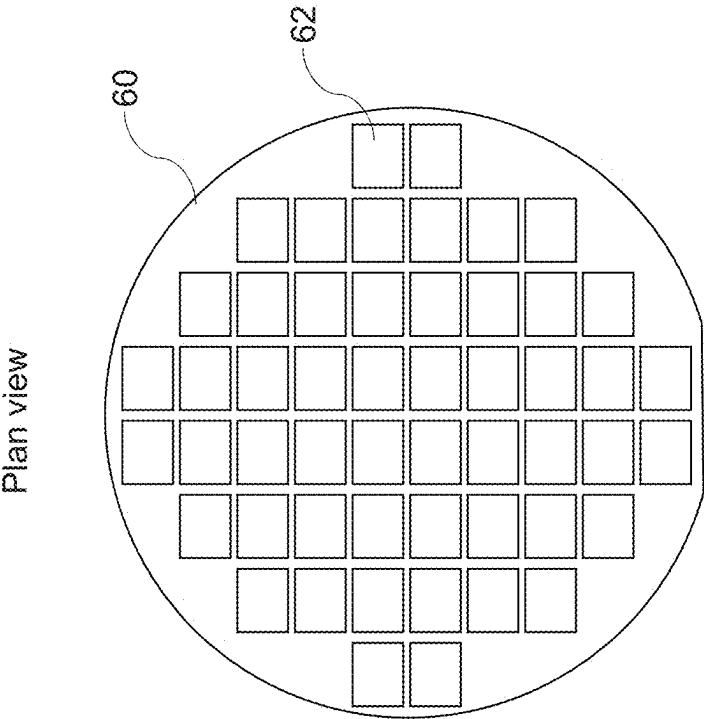


FIG. 12B

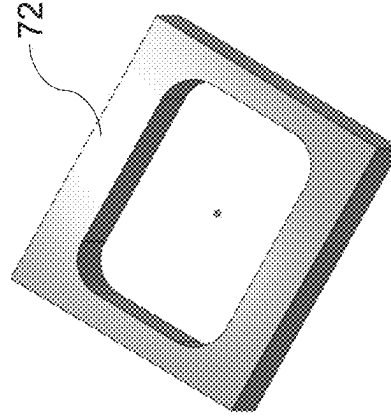


FIG. 12A

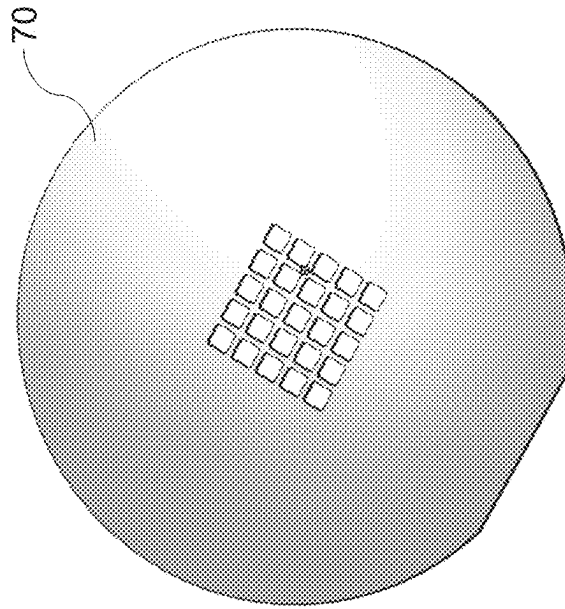


FIG. 13

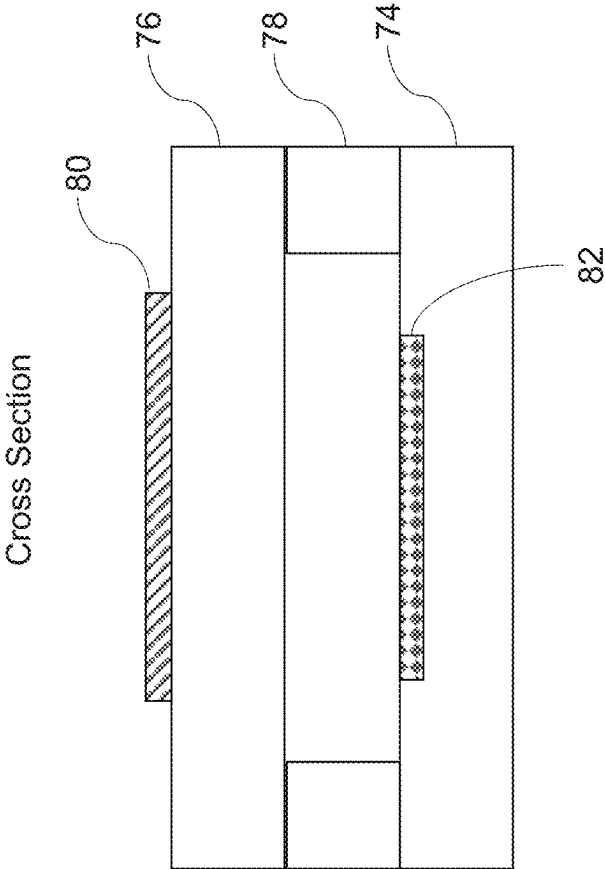


FIG. 14

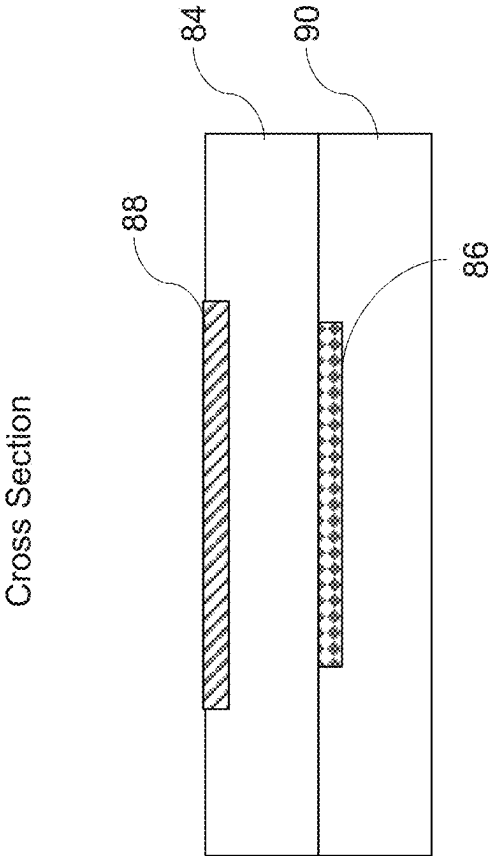
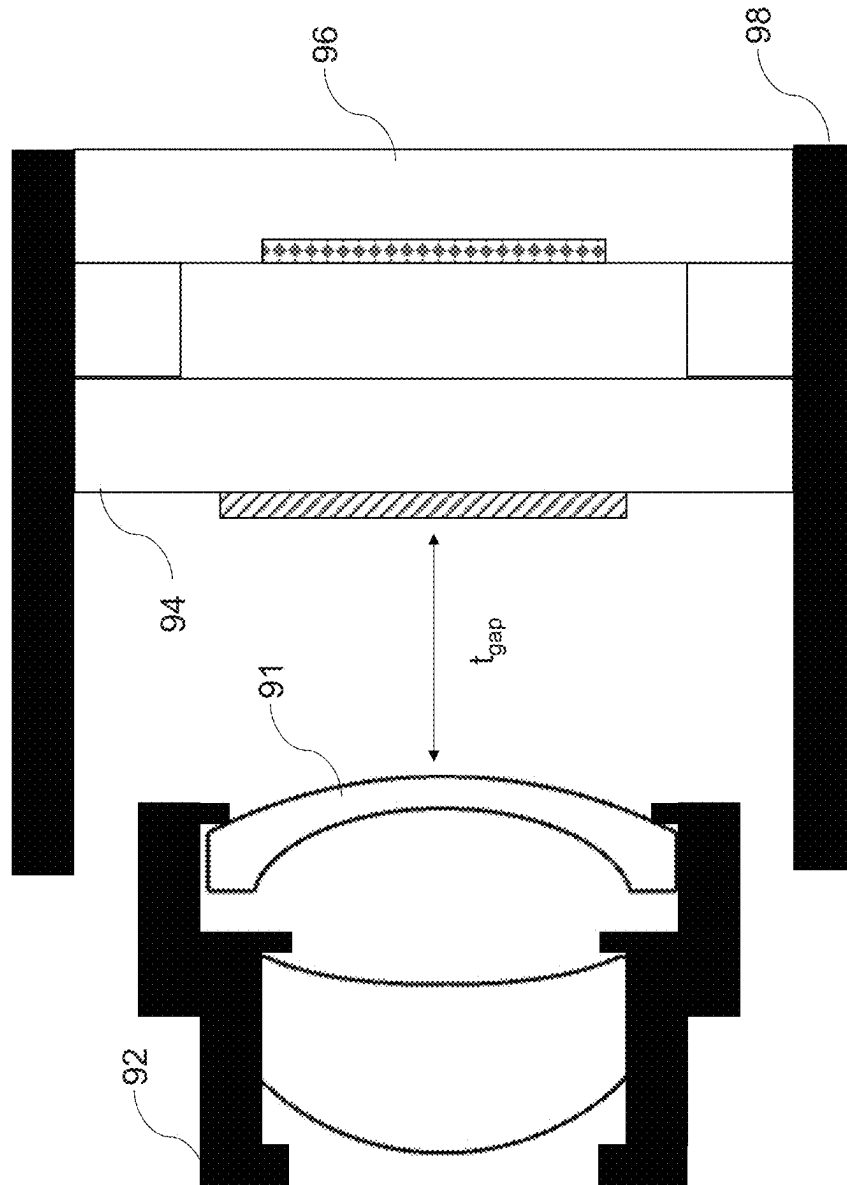


FIG. 15



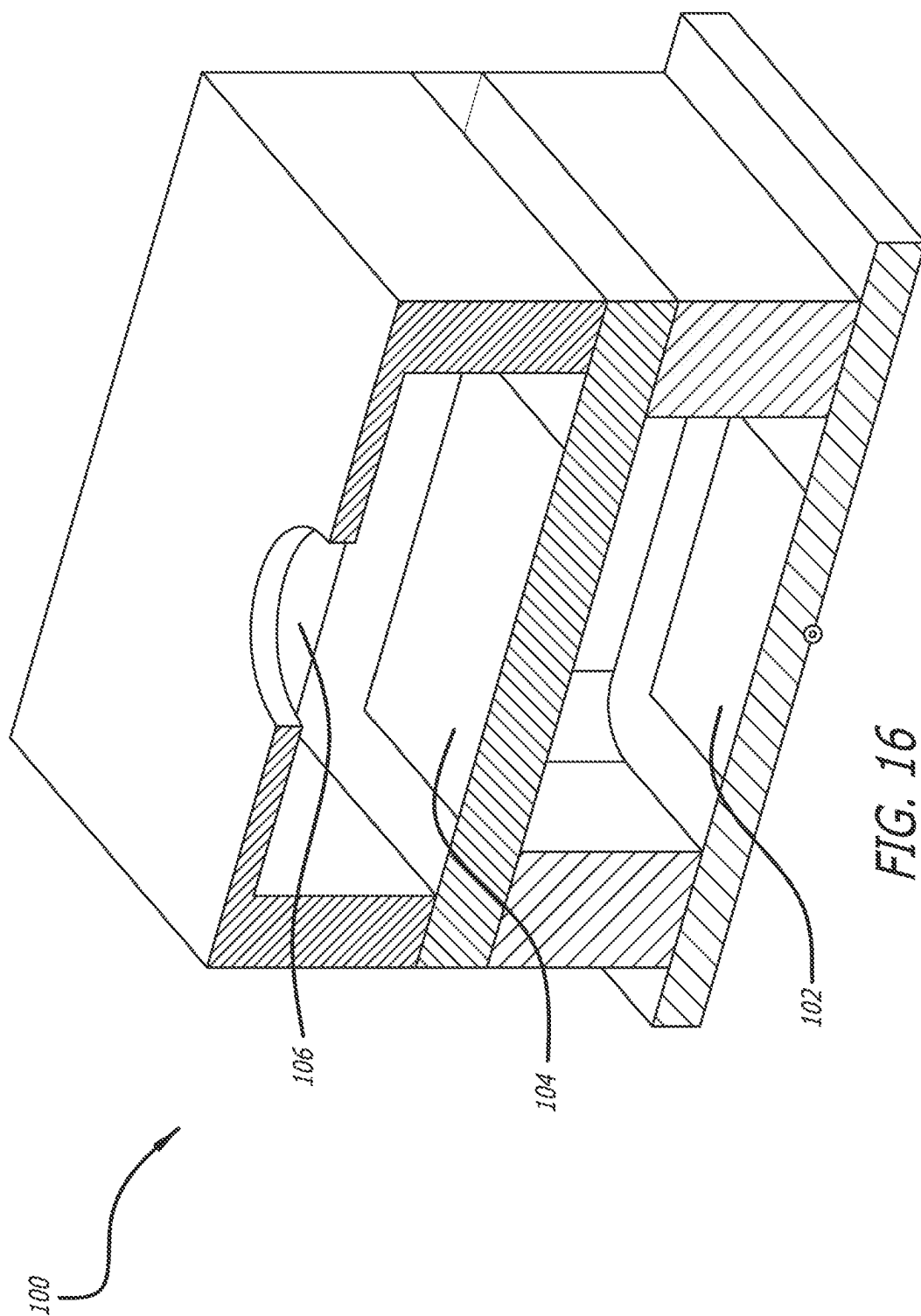


FIG. 17B

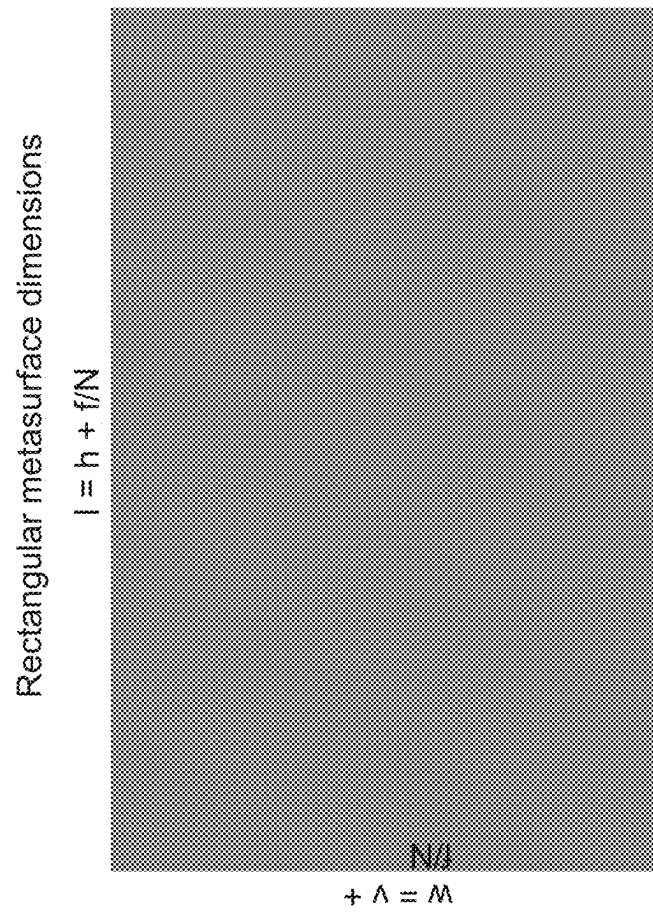
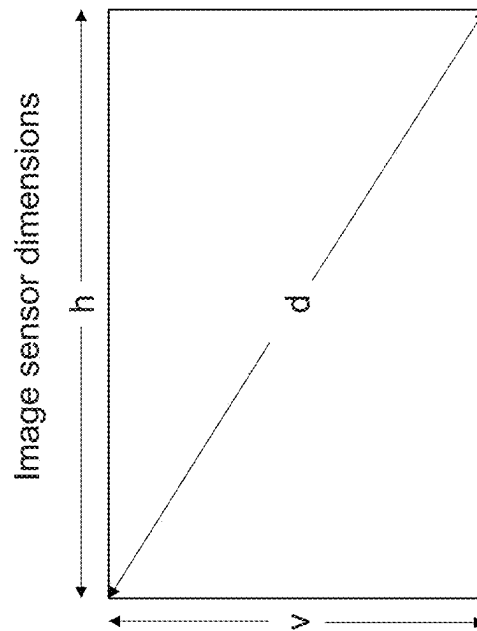


FIG. 17A



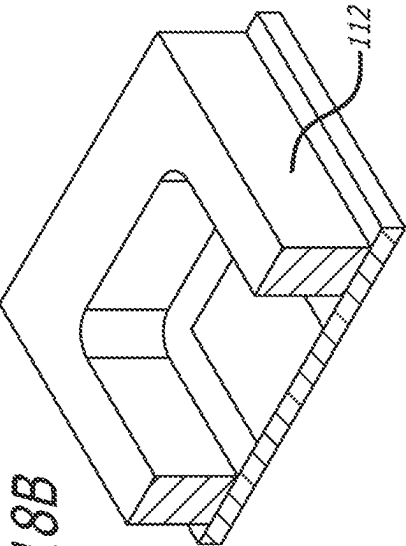


FIG. 18B

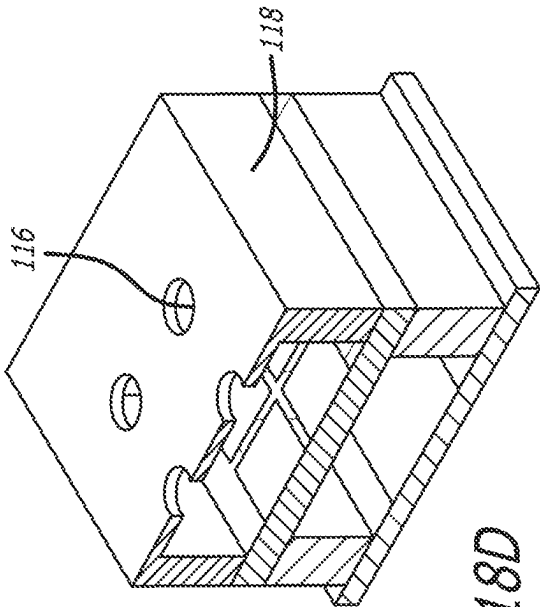


FIG. 18D

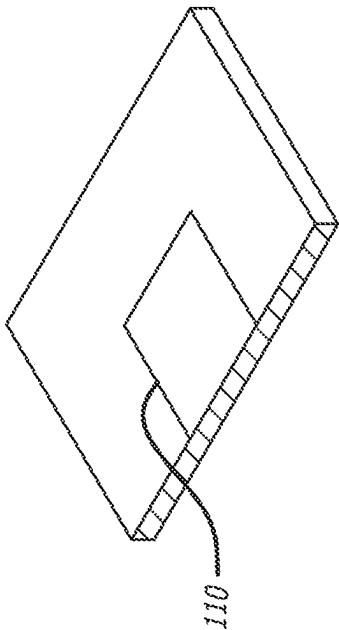


FIG. 18A

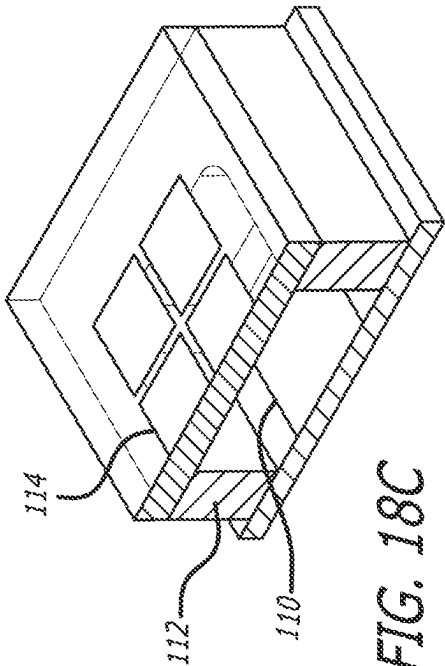


FIG. 18C

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APERTURE-METASURFACE AND HYBRID REFRACTIVE-METASURFACE IMAGING SYSTEMS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/938,823, filed Jul. 24, 2020, which claims priority to U.S. Provisional Patent Application No. 62/878,962, filed Jul. 26, 2019, the disclosures of which are incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

The current disclosure is directed to optical arrangements of metasurface elements, integrated systems incorporating refractive optics, light sources and/or detectors with such metasurface elements, and methods of the manufacture of such optical arrangements and integrated systems.

BACKGROUND OF THE INVENTION

Metasurface elements are diffractive optics in which individual waveguide elements have subwavelength spacing and have a planar profile. Metasurface elements have recently been developed for application in the UV-IR bands (300-10,000 nm). Compared to traditional refractive optics, metasurface elements abruptly introduce phase shifts onto light field. This enables metasurface elements to have thicknesses on the order of the wavelength of light at which they are designed to operate, whereas traditional refractive surfaces have thicknesses that are 10-100 times (or more) larger than the wavelength of light at which they are designed to operate. Additionally, metasurface elements have no variation in thickness in the constituent elements and thus are able to shape light without any curvature, as is required for refractive optics. Compared to traditional diffractive optical elements (DOEs), for example binary diffractive optics, metasurface elements have the ability to impart a range of phase shifts on an incident light field, at a minimum the metasurface elements can have phase shifts between $0-2\pi$ with at least 5 distinct values from that range, whereas binary DOEs are only able to impart two distinct values of phase shift and are often limited to phase shifts of either 0 or 1π . Compared to multi-level DOE's, metasurface elements do not require height variation of its constituent elements along the optical axis, only the in-plane geometries of the metasurface element features vary.

BRIEF SUMMARY OF THE INVENTION

The application is directed to optical arrangements of metasurface elements, integrated systems incorporating light sources and/or detectors with such metasurface elements, and methods of the manufacture of such optical arrangements and integrated systems.

Many embodiments are directed to imaging system including:

- at least one image sensor;
- a substrate layer having a substrate thickness disposed above the at least one image sensor by a first distance, the substrate layer configured to be transparent to a target wavelength of light, the substrate layer having a first surface distal the at least one image sensor and a second surface proximal the at least one image sensor;

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an aperture disposed on the first surface of the substrate and having an aperture opening disposed therein; and a single layer of a plurality of identical or unique nano-structured elements comprising a metasurface disposed on the second surface, such that light impinging on the aperture opening passes through at least a portion of the metasurface such that a specified angular deflection is imposed thereby;

wherein the distance between the aperture and the layer of metasurface elements are separated by a second distance determined by the substrate thickness; and wherein the aperture and the layer of metasurface elements are configured to gather light of a specified operational bandwidth across a specified field of view and shift the incoming light such that it comes to a focus on the at least one image sensor at a zero or near-zero degree chief ray angle.

In still many embodiments, the system further includes a glass cover disposed atop the at least one image sensor.

In yet many the first distance is determined by a spacing layer comprised of one of either a solid-state spacer material or an air gap.

In still yet many embodiments, the field of view is at least ± 30 degrees.

In yet still many embodiments, the system further includes a narrow bandwidth optical filter disposed between the metasurface elements and the at least one image sensor

Various embodiments are directed to an imaging system including:

- at least one image sensor;
- a substrate layer having a substrate thickness, the substrate layer configured to be transparent to a target wavelength of light, the substrate layer having a first surface distal the at least one image sensor and a second surface proximal with the at least one image sensor;
- an aperture disposed above the substrate and having an aperture opening disposed therein; and
- a single layer of a plurality of identical or unique nano-structured elements comprising a metasurface disposed on one of either the first or second surfaces, such that light impinging on the aperture opening passes through at least a portion of the metasurface such that a specified angular deflection is imposed thereby;
- wherein the distance between the aperture and the metasurface layer are separated by a first distance; and
- wherein the aperture and the metasurface layer are configured to gather light of a specified operational bandwidth across a specified field of view and shift the incoming light such that it focuses on the at least one image sensor at a zero or near-zero degree chief ray angle.

In still various embodiment, the system further includes an airgap between the second surface of the substrate and the image sensor.

In yet various embodiments, a spacer layer is disposed within the airgap.

In still yet various embodiments, the metasurface layer is disposed on the first surface.

In still yet various embodiments, the system further includes a narrow bandwidth optical filter disposed on the second surface between the metasurface elements and the at least one image sensor.

In yet still various embodiments, at least a portion of the aperture is interconnected with the first surface.

In still yet various embodiments, the metasurface layer is disposed on the second surface.

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In yet still various embodiments, the image sensor is in contact with the second surface.

In still yet various embodiments, the field of view is at least ± 30 degrees.

Several embodiments are directed to an imaging system including:

at least one image sensor;

substrate layer having a substrate thickness, the substrate layer configured to be transparent to a target wavelength of light, the substrate layer having a first surface distal the at least one image sensor and a second surface proximal with the at least one image sensor;

at least one refractive lens disposed above the substrate and configured to focus impinging light on the first surface of the substrate layer; and

a single layer of a plurality of identical or unique nano-structured elements comprising a metasurface disposed on one of either the first or second surfaces, such that light impinging on the at least one refractive lens passes through at least a portion of the metasurface elements such that an angular deflection is imposed thereby;

wherein the distance between the at least one refractive lens and the layer of metasurface elements are separated by a first distance; and

wherein the refractive lens and the layer of metasurface elements are configured to gather light of a specified operational bandwidth across a specified field of view and shift the incoming light such that it focuses on the at least one image sensor at a zero or near-zero degree chief ray angle.

In still several embodiments, the system further includes an airgap between the second surface of the substrate and the image sensor.

In yet several embodiments, a spacer layer is disposed within the airgap.

In still yet several embodiments, the metasurface layer is disposed on the first surface.

In yet still several embodiments, the system further includes a narrow bandwidth optical filter disposed on the second surface between the metasurface elements and the at least one image sensor.

In still yet several embodiments, at least a portion of at least one of the refractive lenses is interconnected with the first surface.

In yet still several embodiments, the metasurface layer is disposed on the second surface.

In still yet several embodiments, the image sensor is in contact with the second surface.

In yet still several embodiments, the field of view is at least ± 30 degrees.

In still yet several embodiments, the at least one refractive lens is selected from the group consisting of plano-convex, convex-plano, bi-convex, bi-concave, plano-concave, or concave-plano.

In yet still several embodiments, the system includes at least two refractive lenses comprising a convex-concave lens and concave-convex lens.

In still yet several embodiments, the system includes at least three refractive lenses comprising a convex-concave lens, a bi-convex lens and a concave-plano lens.

In various of the above embodiments, at least the imaging sensor and metasurface have rectangular geometries.

In still various of the above embodiments, the at least one refractive lense proximal to the metasurface has a circular geometry.

In yet various of the above embodiments, the image sensor is characterized by a vertical, v , and a horizontal, h ,

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dimension, and wherein the at least one refractive lens is characterized by the f-number of the lens, N , defined as $N=f/D$ where f is the focal length of the optical system and D is the diameter of the lens, and wherein a metalense lens width is given by: $w=v+f/N$, and wherein a metalense length l given by: $l=h+f/N$.

Additional embodiments and features are set forth in part in the description that follows, and in part will become apparent to those skilled in the art upon examination of the specification or may be learned by the practice of the disclosure. A further understanding of the nature and advantages of the present disclosure may be realized by reference to the remaining portions of the specification and the drawings, which forms a part of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The description will be more fully understood with reference to the following figures, which are presented as exemplary embodiments of the invention and should not be construed as a complete recitation of the scope of the invention, wherein:

FIG. 1 provides a schematic illustrating an aperture-metasurface imaging system incorporating a cover glass above the image sensor in accordance with embodiments of the invention.

FIG. 2 provides a schematic illustrating a ray tracing diagram including the chief ray angle at the image sensor plane for the aperture-metasurface imaging system of FIG. 1 in accordance with embodiments of the invention.

FIG. 3 provides a schematic illustrating an aperture-metasurface imaging system with an air gap between the aperture and metasurface in accordance with embodiments of the invention.

FIG. 4 provides a schematic illustrating a ray tracing diagram including the chief ray angle at the image sensor plane for the aperture-metasurface imaging system of FIG. 3 in accordance with embodiments of the invention.

FIG. 5 provides a schematic illustrating an aperture-metasurface imaging system incorporating an air gap above the image sensor and the metasurface layer closer to the object plane in accordance with embodiments of the invention.

FIG. 6 provides a schematic illustrating an aperture-metasurface imaging system incorporating an air gap above the image sensor and the metasurface layer closer to the image plane in accordance with embodiments of the invention.

FIG. 7 provides a schematic illustrating an aperture-metasurface imaging system incorporating an air gap above the image sensor and spacers between the aperture and the metasurface in accordance with embodiments of the invention.

FIG. 8A provides a data graphs showing the relative illumination versus field of view for an aperture-metasurface imaging system in accordance with embodiments of the invention.

FIG. 8B provides a data graph showing the field of view versus degree of distortion for an aperture-metasurface imaging system in accordance with embodiments of the invention.

FIG. 8C provides data graphs showing a modulus transfer function over a field of view of 40 degrees for an aperture-metasurface imaging system in accordance with embodiments of the invention.

FIG. 8D provides an image of a standard test target taken using an aperture-metasurface imaging system in accordance with embodiments of the invention.

FIG. 9 provides a schematic illustrating a single refractive element and metasurface hybrid imaging system in accordance with embodiments of the invention.

FIGS. 10A and 10B provide schematics illustrating multiple refractive element and metasurface hybrid imaging systems in accordance with embodiments of the invention.

FIG. 11A provides a schematic of an image sensor wafer in accordance with embodiments of the invention.

FIG. 11B provides a schematic of an image sensor die in accordance with embodiments of the invention.

FIG. 12A provides a schematic of a spacer wafer in accordance with embodiments of the invention.

FIG. 12B provides a schematic of a spacer in accordance with embodiments of the invention.

FIG. 13 provides a schematic illustrating an integrated hybrid imaging system incorporating refractive elements with metasurface elements and spacers in accordance with embodiments of the invention.

FIG. 14 provides a schematic illustrating an integrated hybrid imaging system incorporating refractive elements with metasurface elements in accordance with embodiments of the invention.

FIG. 15 provides a schematic illustrating a fabrication process for a hybrid refractive element and metasurface imaging system in accordance with embodiments of the invention.

FIG. 16 provides a schematic illustrating an imaging system incorporating a rectangular metasurface lens element in accordance with embodiments of the invention.

FIGS. 17A and 17B provide schematics illustrating the relative dimensions of an imaging sense (FIG. 17A) and a rectangular metasurface lens element (FIG. 17B) in accordance with embodiments of the invention.

FIGS. 18A to 18D provide schematics illustrating optical systems of a N number of apertures over N number of rectangular lenses over a single image sensor in accordance with embodiments of the invention.

DETAILED DESCRIPTION OF THE INVENTION

Turning now to the drawings, hybrid imaging systems incorporating conventional optical elements and metasurface elements with light sources and/or detectors, and methods of the manufacture and operation of such optical arrangements are provided. Many embodiments are directed to systems and methods for integrating apertures with metasurface elements in illumination sources and sensors. Various embodiments, are directed to systems and methods for integrating refractive optics with metasurface elements in illumination sources and sensors.

Embodiments of many optical imaging systems may incorporate a single aperture and single metasurface layer operable to correct for aberrations over a large field of view. Many embodiments of such single aperture and metasurface imaging systems are configured to be telecentric (e.g., having a near 0 degree angle of incidence at image sensor plane) over large field of view such that there is no fall-off in relative illumination over the field of view (e.g., that the intensity from on-axis rays is nearly identical to the intensity at the edge of the field-of-view).

In many embodiments, hybrid refractive optic and metasurface imaging systems may comprise metasurface elements that are free-standing (i.e., not directly integrated with

a specific illuminator or sensor into a system). In some embodiments, the optical system may consist of a single physical component or substrate having a metasurface element disposed on either side thereof. In some embodiments, multiple refractive optics may be combined with at least one metasurface element to make more complex systems.

In embodiments of hybrid aperture or refractive optic and metasurface imaging systems the metasurface may be disposed on a surface of a supporting substrate either facing the aperture or facing the imaging system. In various embodiments airgaps may be disposed between the aperture and metasurface structure and/or between the metasurface substrate and the imaging system. Airgaps between elements may further comprise spacer structures to provide support therefor.

In many embodiment, the metasurface element may be free standing or may be embedded within another material. In various such embodiments, the selection of the embedding material includes the appropriate selection of refractive index and absorption characteristics. In many such embodiments, the embedding material may provide mechanical stability and protection as well as an additional design degree of freedom that enables the metasurface to perform a desired optical function.

In some embodiments, a spacing layer of a defined thickness (e.g., the working distance) may be deposited on the CMOS image sensor, LED, VCSEL, etc., to implement an optical distance appropriate for a desired camera design, illuminator design or optimal system performance. In various such embodiments, the spacing layer material may be organic or inorganic and may have a lower refractive index than the dielectric elements comprising the metasurface. In some such embodiments, the thickness of the spacing layer may be modified to provide appropriate optical spacing for the specific optical system.

Various embodiments are also directed to methods of fabricating hybrid metasurface imaging systems. In some such embodiments, methods are directed to the manufacture of metasurface elements on a wafer incorporating other devices, such as sensors or illuminators, thereby avoiding, in some embodiments, expensive manufacturing processes, such as, for example, the mechanical assembly of small dimension elements, or the active alignment of optics with sensors. In some such embodiments, metasurface elements may be integrated with the sensor (or the illuminator) in a series of operations at a semiconductor fab. In many such embodiments a sequence may include: (i) sensor or illuminator, (ii) optional microlens array/collimator, optional filter, optional spacing layer, optional metasurface element, optional additional spacing layer, optional refractive optic or aperture elements, optional anti-reflection (AR) layer, optional protection layer. In many such embodiments a sequence of elements may include: (i) sensor or illuminator, (ii) optional microlens array/collimator, optional filter, optional spacing layer, optional metasurface element, optional additional spacing layer, and optional refractive element or aperture.

Embodiments for Implementing Aperture/Metasurface Imaging Systems

Typically to form an optical system that is corrected for aberrations over a selected field of view, the system must comprise multiple optical surface or multiple optical elements (e.g., two or more). This is true for both conventional refractive optical systems and metasurface optical systems. Specifically, only optical systems with two or more meta-

surfaces and sufficiently low aberrations over some field of view have been demonstrated. Various embodiments are directed to imaging systems that integrate an aperture and a single metasurface element, that allow for the combined system to achieve high quality imaging over a large field of view, telecentricity (e.g., near 0 degree of incidence at the image sensor plane) over a large field of view, and with no fall-off in relative illumination.

Specifically, such systems may be used in imaging systems such as CMOS cameras, (such as those used in cell phones, computers, tablets etc., for collecting images of a scene of visible light or in the infrared for biometric authentication). These CMOS imaging systems require an increased field-of-view (FOV), independent control of the chief ray angle (CRA) as a function of field height at the CMOS image sensor, and minimal optical distortion of the scene being imaged. These terms will be understood to have a meaning conventional to those skilled in the art. For traditional imaging systems, comprised of refractive lenses, as many as five or six unique lenses must be combined to perform this function. Similarly, in conventional metasurface imaging systems implement multiple metasurface elements to provide enough degrees of freedom to adequately control these parameters (CRA, FOV and minimizing distortion). However, various embodiments show that by combining an aperture with a single metasurface, an imaging system with a wide FOV, controllable distortion and controllable CRA can be realized in accordance with embodiments.

An exemplary embodiment of such a system is illustrated in FIGS. 1 to 7. As shown, in many such embodiments the system (10a to 10d) generally comprises an aperture structure (12a to 12d) disposed a set distance (13a to 13d) away from a metasurface layer (14a to 14d), which itself is set a distance (15a to 15d) away from an image sensor (16a to 16d). As will be described in greater detail below, in such systems it will be understood that the distance between the aperture and metasurface layer, and the distance between the metasurface layer and the imaging system (e.g., the back focal length of the imaging system) may take the form of an air gap or an optically transmissive material (e.g., substrate etc.).

For the purposes of many embodiments, the aperture structure (12a to 12d) comprises a first aperture structure portion (18a to 18d) which is opaque to light at the wavelength of interest and a second aperture structure portion (20a to 20d) that is completely transparent to light at the wavelength of interest over a distance (d_{ap}). In various embodiments such aperture structures impart no optical function (e.g., does not deflect light rays) but rather limits the lateral extent of a light ray bundle entering the imaging system, or otherwise equivalently sets the entrance aperture of the imaging system.

For purposes of many embodiments, the metasurface layer (14a to 14e) generally comprises a plurality of nanostructures (22a to 22e) disposed on a substrate (24a to 24e) defined by a substrate thickness (t_{sub}) which may be formed of any material transparent at the wavelength of interest. In many embodiments of hybrid aperture/metamaterial imaging systems, the metasurface layer is the only functional layer provided that significantly deflects incident light rays to form a focused image (e.g., the metasurface layer operates as an arbitrary phase mask).

Embodiments of nanostructures generally comprise identical or unique three dimension elements (e.g., square, round, triangular, oval, etc.) having feature sizes smaller than the wavelength of light within the specified operational

bandwidth and configured to impose a phase shift on impinging light within the plane of plurality separated by macroscopic distances (distances of 10 or more wavelengths), such that in combination the metasurface layer performs a single optical function. Each individual metasurface in the optical system, may be configured to have some specific 2D phase and transmission function, $\phi(x,y)$ and $t(x,y)$, that it carries out. While in general each metasurface may have a unique distribution of phase and transmission, the nanostructure elements that comprise any metasurface embedded in the same material, with the same base composition and at a specific wavelength are identical. In most practical single wavelength applications, the transmission can be configured to be maximized (near 1) and uniform across the metasurface while the phase can be configured to take on values between 0 and 2π . In summary, according to embodiments, for some wavelength of interest, material system (metasurface material and embedding material), fixed thickness and element spacing, a set of in-plane dimensions of the comprising nanostructures may be configured such that phase delays from 0 to 2π can be imprinted on an incident light field. Thus for different implementations of metasurface designs at the fixed material and wavelength conditions, the only variable from design to design is the distribution of suitable nanostructure elements across the metasurface.

Metasurface layers according to some embodiments may be designed to be freestanding, i.e., the metasurface elements protrude from the end of the substrate with only air gaps separating them, the process is complete at this step. In other embodiments metasurfaces may be further configured to have an AR coating or mechanical protection. In some such embodiments, in order to protect the metasurface and provide improved functionality, the metasurface constituent elements and substrate faces may be coated in some material or layers of material. In embodiments with embedded metasurface elements, the elements, which can be any material with desired optical properties, are embedded in a lower-index of refraction medium. The low-index medium completely encapsulates the metasurfaces and extends some thickness above the metasurface elements. The low-index medium acts as a protective barrier to the metasurface elements (i.e., provide mechanical stability) and provides an additional design degree of freedom for the system that allows for certain properties to be optimized, e.g., overall transmission or efficiency of the metasurface.

Metasurface layers or metasurface systems according to embodiments can be fabricated in mass production using any suitable fabrication techniques, including, for example, lithography, machining, etching, and standard CMOS fabrication techniques, as has been previously described in U.S. patent application Ser. No. 16/120,174, filed Aug. 31, 2018, the disclosure of which is incorporated herein by reference. The metasurface substrate may be any low-index material, e.g., polymer, SiO_2 , glass. The metasurface elements may also be any material which has been optimized for a specific bandwidth, e.g., silicon, TiO_2 , alumina, metal, etc.

The imaging system may take the form of a single monolithic image sensor or a pixel array. Such image sensors and pixel arrays may take any suitable form including, for example, CMOS sensors.

FIG. 1 provides a schematic illustration of an implementation of various embodiments of such hybrid aperture/metamaterial imaging systems. As shown, in many embodiments a substrate layer (24a) transparent at the wavelength of interest and having a thickness (t_{sub}) is provided having an aperture structure (12a) which is opaque to light at the

wavelength of interest and completely transparent to light at that wavelength of interest over a distance, (d_{ap}) disposed on a first side distal to the imager (16a), and a metasurface layer (14a) composed of nanostructures (22a) with equal height disposed on a second side proximal to the imager (16a). In such embodiments the aperture structure (12a) and metasurface layer (14a) are separated by a first distance (13a) defined by the substrate thickness (t_{sub}). Further in such embodiments, the aperture structure (12a) and metasurface layers (14a) may be deposited directly on the substrate (24a) or bonded via adhesive. Between the metasurface layer (14a) and the imager (16a) is disposed a distance (15a) defining the back-focal length formed by an air gap (t_{air}). Although not required, many embodiments of such imaging systems may further comprise an optional cover glass or filter (26) which does not impact the imaging performance of the device but provides other functionality (e.g., either optical or structural).

It will be understood that in such embodiments, the aperture imparts no optical function (does not deflect the light rays) but rather only limits the lateral extent of the light ray bundle that may enter the imaging system, or equivalently sets the entrance aperture or the f/# of the system. Meanwhile, the metasurface layer may comprise the only functional optical layer that significantly deflects the light rays to form a focused image in such embodiments. In some such embodiments, the metasurface layer may act as an arbitrary phase mask, imparting any value from 0 to 2π phase shift on the incident light at an arbitrary radial position of the lens.

Referring to FIG. 2, a ray-tracing diagram through an exemplary embodiment of a system comprising a single aperture (12a) and a single metasurface (14a) combined on a single substrate (24a) in accordance with the embodiment illustrated in FIG. 1 is provided. (Although not described in detail here, it will be understood that these metasurface elements could be fabricated using methods as described herein or in previously cited U.S. patent application No. 16/120,174 using a suitable conformal deposition process, such as, for example, low pressure chemical vapor deposition or atomic layer deposition.) In this exemplary embodiment, the aperture and metasurface elements have been configured such that in combination they are able to form a good image across a wide FOV (± 40 degrees in this example, however, it will be understood that this is not a limiting case). Embodiments of such a single aperture and single metasurface system, as shown, have been surprisingly found to naturally produce focused rays at the image plane that are telecentric (i.e., having 0 degree CRA). In short, while traditional refractive and metasurface designs require complex, many-element systems to realize such telecentric designs, in accordance with embodiments only a single aperture and single metasurface element are needed to achieve similar telecentricity. This telecentricity in turn leads to improved optical properties. In particular, the low (e.g., zero or near zero degree CRA) allows for a narrowing of the bandwidth of the optical filter (26) for narrowband applications. In traditional refractive designs, especially for compact mobile applications, CRAs are typically on the order of 15 degrees to 30 degrees. These larger CRAs in turn require the filter bandwidth to be significantly increased allowing for more ambient light to enter the detector. In narrowband applications (e.g., a near IR VCSEL array), such ambient light can be a persistent noise source. Thus, embodiments of a combined metasurface/filter system such as that shown in FIG. 2 allow for better ambient light performances.

Although FIGS. 1 and 2 provide one arrangement of optical elements for a hybrid aperture/metasurface imaging system, it will be understood that many other arrangements of elements may be realized. For example, FIG. 3 provides a schematic illustration of embodiments of imaging systems in which the position of the airgap and substrate have been exchanged. Such a structure allows for the formation of a thinner imaging system, but entails a more complicated assembly process. In particular, as shown in FIG. 3, such imaging system embodiments comprise a substrate (24b) transparent at the wavelength of interest and having a thickness (t_{sub}) provided having a metasurface layer (14b) composed of nanostructures (22b) with equal height disposed on a first side distal to the imager (16b), and on a second side the imager (16b). In such embodiments, the metasurface layers (14b) and the imager (16b) may be directly bonded via adhesive or other suitable means. Between the metasurface layer (14b) and the imager (16b) is disposed a distance (15b) defining the back-focal length formed by the substrate thickness (t_{sub}). This distance is used as a free parameter to design imaging systems with optimal performance and will change based on the desired f/# or field of view of the imaging system, for example. In such embodiments such imaging systems do not require the optional cover glass or filter used in the embodiment shown in FIG. 1 as the substrate (24b) provides such dual functionality. In such embodiments the aperture structure (12b), which is opaque to light at the wavelength of interest and completely transparent to light at that wavelength of interest over a distance, (d_{ap}), and metasurface layer (14b) are separated by a first distance (13b) defined by and airgap (t_{air}).

Referring to FIG. 4, a ray-tracing diagram through an exemplary embodiment of a system comprising a single aperture (12b), along with a single metasurface (14b) and an imager (16b) combined on a single substrate (24b) in accordance with the embodiment illustrated in FIG. 3 is provided. (Although not described in detail here, it will be understood that these metasurface elements could be fabricated using methods as described herein or in previously cited U.S. patent application Ser. No. 16/120,174 using a suitable conformal deposition process, such as, for example, low pressure chemical vapor deposition or atomic layer deposition.) In this exemplary embodiment, the aperture and metasurface elements have been configured such that in combination they are able to form a good image across a wide FOV (± 40 degrees in this example, however, it will be understood that this is not a limiting case). Embodiments of such a single aperture and single metasurface system, as shown, have been surprisingly found to naturally produce focused rays at the image plane that are telecentric (i.e., having 0 degree CRA).

Although FIGS. 1 and 4 provide arrangements of optical elements for a hybrid aperture/metasurface imaging system in which an element is in direct contact with the image sensor, it will be understood that many other arrangements of elements incorporating spacers disposed between the image sensor and the substrate supporting the metasurface layer may be realized. For example, FIG. 5 provides a schematic illustration of embodiments of imaging systems in which a second airgap (28) is disposed within the imaging system.

In particular, as shown in FIG. 5, such imaging system embodiments comprise a substrate (24c) transparent at the wavelength of interest and having a thickness (t_{sub}) provided having a metasurface layer (14c) composed of nanostructures (22c) with equal height disposed on a first side distal to the imager (16c), and a second airgap (28) disposed

between a second side of the metasurface layer proximal to the imager (16c). One advantage of such embodiments incorporating an air gap is that the light rays proceed through the system at higher angle, as compared to the embodiment shown in FIG. 1, for example, thus allowing for a decrease in the overall form factor of the metasurface optical system. In addition, the gap between the metasurface substrate and image sensor alloys for the introduction of other optical elements, including, for example, microlens arrays or optical color filters to improve optical function of the imaging system.

In such embodiments, the metasurface layers (14c) and the imager (16c) may be deposited directly on the substrate (24c) or bonded via adhesive. Such embodiments may also comprise suitable spacers (30) to support the substrate (24c) and maintain the distance between the substrate and the image sensor (16c). Between the metasurface layer (14c) and the imager (16c) is disposed a distance (15c) defining the back-focal length formed by the combination of the substrate thickness (t_{sub}) and a spacer height (t_{spacer}). The spacers (30) can be either fixed to the image sensor (16c) and substrate layer (24c), leading to a fixed distance for (t_{spacer}) or the substrate can be placed into a standard optical barrel and (t_{spacer}) can be adjustable post assembly. Such embodiments allow for the surface (34) of the substrate (24c) proximal to the image sensor (16c) to remain unpatterned, allowing for the direct integration of an optional optical filter thereon.

Although one embodiment of a configuration of a hybrid aperture/metamaterial incorporating an airgap above the image sensor has been described, as shown in FIG. 6, in various embodiments the metasurface layer (14d) may also be disposed on the surface of the substrate (24d) proximal to the image sensor (16d) facing the air gap (32) supported by spacers (30'). Such an implementation allows for the protection of the metasurface elements from environmental contamination. Additionally, such embodiments allow for the surface (34') of the metasurface substrate (24d) distal to the image sensor (16d) to remain unpatterned, allowing for the direct integration of an optional optical filter on the substrate. Again, in such embodiments the spacer (30') can be either fixed to the image sensor (16d) and substrate (24d), leading to a fixed distance for (t_{spacer}) or the substrate (24d) can be placed into a standard optical barrel and (t_{spacer}) can be adjustable post assembly.

Accordingly, embodiments illustrated in FIGS. 5 and 6 illustrate that the metasurface elements may be arranged to face inward or outward with respect to the air gap between the substrate and image sensor. Production of the metasurface system illustrated in FIGS. 5 and 6 can follow processes described, for example, in U.S. patent application Ser. No. 16/120,174. The spacer layers may be any low-index material, e.g., polymer, SiO_2 , glass.

Although embodiments of hybrid aperture/metamaterial imaging systems incorporating an airgap between the aperture and metasurface substrate have been shown in FIGS. 3, 5 and 6, these embodiments would require a separate supporting structure to secure the aperture and ensure the aperture distance (t_{air}) remains constant. However, the aperture structure (12e) may also be attached directly to the substrate layer (24e). An exemplary embodiment of such an imaging system is illustrated in FIG. 7. As shown, in this exemplary embodiment the top aperture (12e) has an aperture body (36) with width $d_{ap,top}$ that sets the entrance aperture of the system, has an aperture offset from the substrate by a distance (t_{ap}), and that is angled with a minimum angle set by the half field of view of the imaging

system to a width of the aperture after a distance along the optical axis set by (t_{ap}) is set by the width of the metasurface layer and is given by ($d_{ap,bottom}$). Although the embodiment shown in FIG. 7 depicts a system in which the metasurface layer (14e) is disposed on a surface of the substrate (24e) distal from the image sensor (16e), it will be understood that the metasurface layer may also be disposed on the surface (34') of the substrate proximal to the image sensor. In addition, it will be understood that while the exemplary embodiment incorporates a spacer structure (30') and airgap between the substrate (24e) and image sensor (16e), embodiments may omit this element and mount the substrate directly atop the image sensor.

An attribute of embodiments of such telecentric designs is that the metasurface imaging system provides a more uniform illumination at the image sensor (referred to by those in the art as "relative illumination"). A data plot of relative illumination for an exemplary system according to embodiments is provided in FIG. 8A, and demonstrates that relative illumination for the aperture/metamaterial imaging system is maintained at 100% across the entire field of view, which is a substantial improvement over conventional systems, which can have a difference between center and edge of 50% or more. Accordingly, embodiments of imaging systems are able to collect more total illumination across the full field of view. Embodiments of the metasurface system also provide an additional design variation with respect to traditional refractive lens systems. Typical CMOS image sensors (CIS) require a microlens to be associated with each pixel. Because there is a large variation in CRA across a given sensor plane that is inherent to refractive optical systems, the microlens array on the CIS also requires a complex CRA specification. However, in embodiments of metasurface systems as described herein, the CRA of the microlens array may be configured to be a constant 0 degrees across the CIS allowing for greater simplicity in the design and fabrication of the microlens array. Alternatively, in certain implementations the microlens array may be completely removed from the CIS, saving a process step in CIS production. Furthermore, such aperture metasurface systems with 0 CRA allow one to limit a persistent problem in traditional imaging systems known by those in the art as "pixel cross-talk". Traditional refractive systems that send light into the image sensor are prone to light coupling into neighboring pixels, which adds noise to the system.

Conventional metasurface systems may be configured with multiple metasurface layers to control for FOV and distortion. The introduction of the additional metasurface element or elements, allow for the realization of a separate arbitrary phase profile, provides more degrees of freedom to control the path of the light rays as compared to a typical system comprised of an equivalent number refractive elements. In embodiments of the current imaging system where a single metasurface layer is used it is not possible to control CRA and correct for grid distortion simultaneously. As a result, a certain amount of grid distortion is inevitable. For example, FIG. 8B provides a data plot illustrating distortion as a function of field at the CMOS image sensor for an imaging system based on the embodiment shown in FIG. 1. FIG. 8C shows a plot of the modulus transfer function over the field of view for an embodiment of an imaging system as shown in FIG. 1. FIG. 8D provides a standard test image illustrating the grid distortion for an imaging system based on the embodiment shown in FIGS. 8A to 8C. As shown, at the edge of the field of view the quality drops off. However, embodiments of the invention show that all of the wave-

lengths of light are tightly bundled indicating that the distortion can be corrected simply using known image processing software.

Embodiments Implementing Hybrid Refractive and Metasurface Elements

Although embodiments incorporating apertures and single metasurface layers have been described, it will be understood that embodiments are also directed to hybrid systems of metasurface elements incorporating refractive lens elements. FIG. 9 provides a schematic illustration of an implementation of various embodiments of such hybrid refractive lens/metasurface imaging systems. In many embodiments the hybrid imaging surface consists of at least one of each of the following: a refractive lens with one or more surface of curvature and a metasurface layer on a substrate where all of the elements comprising the metasurface are the same height. More generally, the hybrid optical system may be comprised of any number of refractive elements and multiple metasurface layers. In certain embodiments having the metasurface element on the substrate layer that is the final component prior to the image sensor provides the specific advantage of creating a so-called image-space telecentric imaging system. In addition, the substrate upon which the metasurface layer is formed may have a set of filter layers deposited. As described in embodiments incorporating a filter, the metasurface layer is closer to the object plane of the imaging system while the near infrared filter is closer to the image sensor.

An exemplary embodiment of a hybrid refractive element/met surface imaging system is illustrated in FIG. 9. In many embodiments the system includes at least one refractive optic (38) disposed a set distance away from a substrate layer (40) transparent at the wavelength of interest and having a thickness (t_{sub}) having a metasurface layer (42) composed of nanostructures (44) with equal height disposed on a first surface (45) thereof that is distal to the image sensor (46), and an optional optical filter (48) disposed on a second side (47) of the substrate proximal to the imager.

Although the embodiment shown in FIG. 9 illustrates a hybrid system having a single refractive optic and a single metasurface layer, it will be understood that embodiments may also incorporate other arrangements of refractive optics. Figures illustrating such metasurface/refractive hybrid systems are provided in FIGS. 10A and 10B. Specifically, FIG. 10A illustrates a hybrid system comprising two refractive elements (50 & 51) and one metasurface layer (52) system where the two refractive elements are convex-concave and concave-convex. FIG. 10B illustrates a hybrid system comprising three refractive elements (54, 55, 56) and one metasurface element (58) where the three refractive elements are convex-concave, bi-convex and concave-plano, respectively.

With regard to the refractive optical elements in the hybrid systems described above, i.e., those elements preceding the metasurface layer, it will be understood that the surface curvature of these elements may take on any positive, negative or infinite values. Accordingly, although specific arrangements of refractive elements are shown in the figures, the refractive elements may take any suitable form and combination for the specific application, including for example, plano-convex, convex-plano, bi-convex, bi-concave, plano-concave, or concave-plano.

Embodiments Implementing Image Sensor Wafers

Imaging systems known in the art typically consist exclusively of traditional refractive lenses (glass or plastic mate-

rials with at least one curved surface). According to embodiments a single metasurface layer is disposed in combination with one or more, curved refractive lens. Surprisingly, it has been found that the inclusion of the single metasurface layer turns the optical system telecentric. Specifically, in many such embodiments, the inclusion of the metasurface element as the final element before the image plane makes the system telecentric.

Embodiments of metasurface layers may be integrated with the CMOS image sensor (CIS) as the cover glass and filter while the refractive optics can be assembled in barrels as is done conventionally in optical imaging modules. In certain embodiments the glass upon which the metasurface layer is fabricated, in addition to be a cover glass for the CIS, can have been previously deposited with dielectric layers and act as a near infrared bandpass or long pass filter. Such embodiments provide a single component with function in the optical imaging process as well as to eliminate unwanted wavelengths from being incident on the image sensor.

While the above embodiments have focused on hybrid metasurface imaging systems with a single sensor element, e.g., as shown in FIGS. 1 to 10B, metasurface elements can also be integrated with an image sensor wafer containing a plurality of image sensor dies. A figure illustrating a schematic for an image sensor wafer is shown in FIG. 11A. As shown, embodiments may comprise an image sensor wafer (60) comprising a set of image sensor dies (62), while this is shown in a periodically spaced 2D array, it will be understood that the array need not be periodically spaced. As shown in FIG. 11B, in turn, each sensor die (62) comprises an image sensor active area (64). Although each image sensor may be identical, in many embodiments the character of each sensor may be generally unique. Because each imager comprising array may have unique properties, it may also be advantageous to have an array of metasurface elements, each with uniquely designed properties.

It will be understood that only the active area (64) need be available for imaging. The region outside of the image sensor active area (66) may be used to attach lenses or spacers. Moreover, in various embodiments the metasurface substrate may be offset from the image array by a spacer. Exemplary embodiments of a spacer wafer (70) comprising a plurality of spacer openings suitable for attachment to a sensor wafer, and a spacer die (72) suitable for attachment to an image sensor or lens die are illustrated in FIGS. 12A and 12B, respectively. The design and thickness of the spacer layer will depend on the specific configuration, but in many embodiments the thickness is configured to allow the light from the illumination sources to diverge sufficiently before interacting with image sensor. Again, the function of each metasurface element in the array may be generally unique, and may be patterned on top of each individual image sensor in the array, utilizing any suitable technique outlined for example in U.S. patent application Ser. No. 16/120,174. For example, a metasurface may be fabricated directly on each individual image sensor in the array or a suitable dielectric spacer may be deposited on the image sensor followed by the integration of the metasurface on top of the combined dielectric layer and image sensor. In such embodiments, the metasurfaces may provide a particular radiation pattern for each image sensor and the entire system (image sensor properties, geometrical parameters and metasurface-enabled radiation pattern) can be iteratively optimized for a specific set of performance parameters.

In various other embodiments, a dielectric material, with an index of refraction lower than that of the constituent image sensor material may be deposited and planarized such

that a single metasurface can be patterned on top of the dielectric material. This contrasts with embodiments where each image sensor in the array has a unique metasurface patterned on its facet. Again, in such embodiments the combined system may be optimized to achieve a desired performance. Finally, in all of the above embodiments, integration of a metasurface with an image sensor array may be accomplished using wafer level optics processes. In such embodiments, the spacer layer may be air rather than a solid dielectric, illustrations of exemplary embodiments of such devices are shown in FIGS. 13 and 14.

Specifically, FIG. 13 shows a schematic of an embodiment comprising an image sensor die (74) and a lens die (76) separated by a spacer (78). In such embodiments, the spacer (78) controls the distance between the metasurface area (80) and the image sensor active area (82). The spacer in such embodiments can be attached to the image sensor die or image sensor wafer then the lens can be attached to the spacer. Alternatively the spacer can be first attached to the lens, then the sensor attached to the spacer. In such embodiments, the spacer can be attached using adhesives (e.g. UV-cured epoxy or thermal cured epoxy), solders, or fusion bond.

FIG. 14, illustrates and exemplary embodiment that excludes a spacer. In such embodiments, the thickness of the lens die (84) determines the distance between the image sensor active area (86) and the metasurface area (88). The image sensor die (90) and lens die (84) can be directly attached, using adhesive, solder, fusion bond, bump bond, etc. Alternatively the image sensor wafer and lens wafer can be attached directly at the wafer level as previously described.

As shown in FIG. 15, in certain embodiments, the refractive lenses (91) of hybrid systems may first be assembled in a lens barrel (92), as is already known in the art. The metasurface element (94) can then be combined with the CMOS image sensor element (96) as described above with respect to FIGS. 13 and 14, above. These two sub components are then assembled together to form the final system. In such embodiments, the refractive lenses may be configured to thread into a housing (98) such that the distance between the refractive elements and metasurface (t_{gap}) may be adjusted.

Embodiments Implementing

As is well known in the arts, the image of a scene formed by a circular, radially-symmetric lens or system of circular, radially symmetric lenses will also be a circle. As a result, the geometry of the formed image is often referred to as the image circle of the lens. In modern photography, however, the medium recording the image (CMOS image sensor, e.g.) often has a rectangular shape. In the camera design, the image circle of the lens is designed such that the diameter of the image circle, D_{image} , is at least as large as the diagonal of the image sensor, d . However, since the image sensor is rectangular, only a portion of the image circle falls actually onto the image sensor. Thus, much of the lens area where light from a scene is incident is not used in the final formation of an image from the camera system.

In a traditional injection molded plastic refractive lens, the shape of the lens is kept ideally circular. The circular shape is used because from a manufacturing perspective circular shapes and radially-symmetric lenses are the easiest to achieve and most repeatable in production. Furthermore, there is minimal cost increase of making a circular lens with larger area rather than a rectangular lens with a smaller area.

Thus, conventional cameras use circular lenses and only a portion of the light impinging upon the full circular lens is collected by the rectangular image sensor. These portions of the circular lens where light is incident but does not contribute to light falling on the image sensor are not used in the final image formation.

For the metasurface lenses according to various embodiments, the shape of the lens can be engineered such that the image it forms is uniquely matched with a specific image sensor dimension. In contrast to conventional refractive lenses, in many such embodiments, the lens shape is configured to be no longer circular and the lens to no longer form an image circle. In various embodiments the metasurface lens is formed in a rectangular configuration with specific dimensions and thus the formed scene image is also rectangular. In the ideal case of such a design embodiment, all of the light impinging on the rectangular metasurface lens falls onto the image sensor. Embodiments of metasurfaces having rectangular lenses break the radial symmetry and therefore the image that the lens forms is no longer circular or radially symmetric.

Accordingly, in many embodiments of lens systems the metasurface lens element may be formed in a rectangular configuration. Advantages of such rectangular or not circular lenses include: limiting the total area of the lens, eliminating portions of the lens that would otherwise have light impinge upon it that does not subsequently form an image on the image sensor, and simplifying post processing of lens wafers. Particular embodiments of rectangular metasurface lenses and the imaging systems are described here.

Many embodiments of metasurface lens systems having non-circular configurations have the following commonalities: an entrance aperture which is the stop of the lens system and a metasurface lens which is the final, active optical surface before the image sensor plane. In such embodiments, the entrance aperture (and optical stop) can be circular in cross-section, as it would be in a traditional optical system, while the metasurface lens can be patterned as a rectangle or other arbitrary shape.

FIG. 16 provides an illustration of an exemplary embodiment of a clear aperture metasurface, where the imaging system (100) is comprised of a rectangular image sensor (102) offset from a single rectangular metasurface (104), and a circular entrance aperture (106) offset from the rectangular metasurface lens (104). In embodiments implementing hybrid metasurface refractive systems, the optical systems are comprised of an entrance aperture, at least one refractive lens and a metasurface lens which is the last optical lens element in the system before the image sensor. In these embodiments of such hybrid systems the entrance aperture and the at least one refractive lens can still have a circular or radially symmetric cross section. Thus, embodiments of such hybrid systems would be comprised of a circular aperture (106) offset from at least one circular refractive lens (not shown) which is then offset from the rectangular metasurface lens (104), as shown in FIG. 16. Again, in such embodiments the metasurface lens has rectangular dimensions which are configured to match a specific image sensor. Although example here describe implementations where the metasurface lens element alone has a rectangular cross section, this need not be a limiting case. For example, the entrance aperture (and stop of the optical system) could also be rectangular or the at least one refractive lens could also be rectangular as long as at least the metasurface lens element, which is the last lens element before the image sensor, has a rectangular cross section.

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In certain cases, the dimensions of the rectangular lens in the system can be fully characterized by the dimensions of the image sensor in the system and the specifications of the lens. Specifically, as shown in FIG. 17A, an image sensor is characterized by its vertical and horizontal dimensions, v and h , respectively. A lens system is characterized most generally by the f -number of the lens, N , defined as $N=f/D$ where f is the focal length of the optical system and D is the diameter of the lens. Accordingly, as shown in FIG. 17B, the desired rectangular metalens lens width can be shown to be $w=v+f/N$ and the length can be shown to be $l=h+f/N$. Such definitions of the lens dimensions will result in an image that almost perfectly fills the sensor geometry. In practice, it is desirable to have the image formed by the imaging system be slightly larger than the image sensor dimensions. This oversizing of the image allows for greater tolerance in the final assembly of the lens system. A typical oversizing range may be to take the nominal rectangular lens dimension, as given above, and increase each dimension by 40 microns.

While the above example specifies a single circle aperture coupled with a rectangular lens, in other embodiments an optical system of a N number of apertures over N number of rectangular lenses over a single image sensor may also be provided. For example, the circular apertures and rectangular metasurface lenses may be arrayed in an 2×2 grid over a single image sensor. An example of such a system is shown in FIGS. 18A to 18D. FIG. 18A shows a single image sensor die (110) around which a single spacer (112) can be placed as shown in FIG. 18B. The spacer sets the distance between the rectangular metasurfaces (114), shown in FIG. 18C and the image sensor (110). The last element in the assembly contains a set of circular apertures (116) disposed in association with an additional spacing layer (118) to set the distance between the rectangular metasurface and the circular aperture. A cut away of the full assembly is shown in FIG. 18D.

Although FIGS. 18A to FIG. 18D exemplify a 2×2 array of circular apertures over a rectangular metasurface, in general, the grid of circular apertures and rectangular metasurface lenses may be arrayed in any fashion, either symmetric or asymmetric, e.g., 3×3 or 5×2 . Additionally, each individual rectangular metasurface and circular aperture comprising the system may have unique dimensions relative to the other aperture metasurface pairs in the system. For example, each aperture diameter in the array can generally be unique or each rectangular lens may be unique. However, the distance between the metasurface lens and the image sensor and the metasurface lens and the circular apertures is generally fixed for the entire system. Changing the mechanical parameters, e.g., aperture size, for each individual component of the array allows the optical properties of each camera in the array to be unique. For example, according to embodiments each sub camera can have a unique $f/\#$, field-of-view, resolution, etc.

DOCTRINE OF EQUIVALENTS

Accordingly, although the present invention has been described in certain specific aspects, many additional modifications and variations would be apparent to those skilled in the art. It is therefore to be understood that the present invention may be practiced otherwise than specifically described. Thus, embodiments of the present invention should be considered in all respects as illustrative and not restrictive.

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What is claimed is:

1. An imaging system comprising:

at least one image sensor;

a substrate layer having a substrate thickness disposed above the at least one image sensor by a first distance, wherein the substrate layer:

is configured to be transparent to a target wavelength of light; and

has a first surface distal the at least one image sensor and a second surface proximal the at least one image sensor;

an aperture disposed on the first surface of the substrate layer and having an aperture opening disposed therein; and

a single layer of a plurality of identical or unique nano-structured elements comprising a metasurface disposed on the second surface, wherein:

light impinging on the aperture opening passes through at least a portion of the metasurface such that a specified angular deflection is imposed thereby;

the metasurface is separated from the at least one image sensor by the first distance;

the aperture and the metasurface are separated by a second distance which is the substrate thickness;

a combination of the second distance and the first distance is configured to:

gather light of a specified operational bandwidth across a specified field of view, and

shift incoming light such that it comes to a focus on the at least one image sensor at a zero or near-zero degree chief ray angle; and

a given image sensor of the at least one image sensor and the metasurface each comprise rectangular geometries in plan view.

2. The imaging system of claim 1, further comprising a glass cover disposed atop the at least one image sensor.

3. The imaging system of claim 1, wherein the first distance is determined by a spacing layer comprised of one of either a solid-state spacer material or an air gap.

4. The imaging system of claim 1, wherein the specified field of view is at least ± 30 degrees.

5. The imaging system of claim 1, further comprising a narrow bandwidth optical filter disposed between the metasurface and the at least one image sensor.

6. The imaging system of claim 1, further comprising at least one refractive optic positioned adjacent to the first surface of the substrate layer.

7. The imaging system of claim 1, wherein the aperture opening does not deflect light rays.

8. The imaging system of claim 1, further comprising an anti-reflective coating on the metasurface, the first surface, and/or the second surface.

9. The imaging system of claim 1, wherein the at least one image sensor comprises multiple image sensor dies periodically spaced in a 2D array.

10. An imaging system comprising:

at least one image sensor;

a substrate layer having a substrate thickness, the substrate layer configured to be transparent to a target wavelength of light, the substrate layer having a first surface distal the at least one image sensor and a second surface proximal with the at least one image sensor; an aperture disposed above the substrate layer and having an aperture opening disposed therein; and

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a single layer of a plurality of identical or unique nano-structured elements comprising a metasurface disposed on one of either the first surface or the second surface, wherein:

light impinging on the aperture opening passes through at least a portion of the metasurface such that a specified angular deflection is imposed thereby;

the metasurface is separated from the at least one image sensor by a first distance;

the aperture and the metasurface are separated by a second distance;

a combination of the second distance and the first distance is configured to:

gather light of a specified operational bandwidth across a specified field of view; and

shift incoming light such that it focuses on the at least one image sensor at a zero or near-zero degree chief ray angle; and

a given image sensor of the at least one image sensor and the metasurface each comprise rectangular geometries in plan view.

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11. The imaging system of claim 10, further comprising an airgap between the second surface of the substrate layer and the at least one image sensor.

12. The imaging system of claim 11, wherein a spacer layer is disposed within the airgap.

13. The imaging system of claim 11, wherein the metasurface is disposed on the first surface.

14. The imaging system of claim 13, further comprising a narrow bandwidth optical filter disposed on the second surface between the metasurface and the at least one image sensor.

15. The imaging system of claim 13, wherein at least a portion of the aperture is interconnected with the first surface.

16. The imaging system of claim 11, wherein the metasurface is disposed on the second surface.

17. The imaging system of claim 10, wherein the given image sensor is in contact with the second surface.

18. The imaging system of claim 10, wherein the specified field of view is at least ± 30 degrees.

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